Income and Energy Tax Pass-through: Evidence from Gas Stations

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Abstract

The distributional impact of energy taxation depends on not just relative consumption but also relative price changes. I illustrate this by estimating the pass-through of staggered tax changes in Spain's retail automotive fuel market, using a new difference-in-differences method (de Chaisemartin and D'Haultfoeuille, 2024a) that avoids key issues with two-way fixed effects models. I find that gas stations passed through regional diesel tax increases from 2010-2013 at a rate of roughly 92 percent overall, and at significantly higher rates in municipalities with higher median incomes. Stations in the highest four income quintiles have pass-through rates that are 7-11 percentage points larger than stations in the bottom income quintile and stations in the smallest municipalities lacking income data. Branding and market structure do not fully explain the pass-through/income relationship. The sign and magnitude of this relationship are consistent with a partial offsetting of overall regressive consumer surplus impacts of the tax.

Keywords: pass-through; distributional equity; fuel taxes; energy policy
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Introduction

Taxation is a fundamental instrument of energy and environmental policy, used widely in transportation, electricity, and other emissions-intensive markets. Economists tend to support such taxes due to the gains in total social welfare from "pricing in" externalities. The *distribution* of welfare impacts of taxation, however, is not guaranteed to be progressive. A common first step in assessing the fairness of energy taxation (or that of any other good) is to examine how household expenditure on the taxed form(s) of energy, as a proportion of its total expenditure, varies across the wealth (or proxy thereof) distribution. Gross of any redistribution of tax revenues, energy taxes are then judged as regressive when the poor devote a larger portion of their total expenditure to the taxed energy relative to to the rich, and progressive when the poor devote a *smaller* portion. Empirically, this relationship varies across countries and types of energy: for instance, globally, electricity taxes appear uniformly regressive, while transportation fuel taxes sometimes are progressive (Flues, F. and Thomas, A., 2015; Pizer and Sexton, 2020).

This method for judging fairness relies primarily on "relative quantities" – that is, how much of the taxed good in question is consumed by the rich relative to the poor. Yet it is not just relative quantities that dictate regressivity; it is also relative price impacts. The first-order approximation of a tax's effect on consumer surplus is a function of both consumption *and* price changes. That is, *pass-through* of the tax – from its point of statutory incidence, to prices paid by households – has the potential to be a significant determinant of how the burden of energy taxes is distributed across the wealth spectrum. Distributional welfare analyses of commodity taxation in general assume that pass-through of taxes is uniform across households (Gruber and Koszegi, 2004; Bento et al., 2009; Horowitz et al., 2017). But recent research has uncovered an abundance of heterogeneity in energy tax pass-through rates; in automotive fuel markets, for example, such rates have been found to vary as a function of market power (Doyle Jr. and Samphantharak, 2008; Genakos and Pagliero, 2022), vertical structure (Bajo-Buenestado and Borrella-Mas, 2022), and supply elasticity (Marion and Muehlegger, 2011; Kilian and Zhou, 2024). Only one existing study of energy tax pass-through focuses on its relationship with wealth (Harju et al., 2022), and it finds higher pass-through rates of a Finnish auto fuel tax hike in zip codes with lower average income.

This paper provides an alternative data point on the empirical relationship between energy tax pass-through and income, by studying tax changes in Spain's retail automotive fuel market. I collect daily prices of auto diesel – the dominant auto fuel in Spain – at nearly 10,000 gas stations across the country between 2007 and 2013, made available through an informational mandate unveiled in January 2007 by the Spanish government. I combine these data with panel-varying regional fuel taxes and gas station attributes, including municipality-level income data, and use a difference-in-differences (DD) strategy to estimate pass-through nationally and in subgroups of interest.

From an estimation standpoint, the setting is similar to those of other fuel tax pass-through studies (Alm et al., 2009; Marion and Muehlegger, 2011; Kilian and Zhou, 2024) and common in observational economics studies generally (see, e.g., de Chaisemartin and D'Haultfoeuille, 2024a): the "treatment" is a series of (fourteen) staggered, area-specific tax changes of differing magnitudes. A growing econometrics literature (de Chaisemartin and D'Haultfoeuille, 2020; Sun and Abraham, 2021; Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021; Borusyak et al., 2024) has shown that the conventional identification strategy in this context among economists, two-way fixed effects (TWFE) regression, is poorly suited to the task of treatment effect estimation when such effects vary over time and across space. Among several new related methods for credible estimation of treatment effects via "staggered difference-in-differences", the method of de Chaisemartin and D'Haultfoeuille (2024a) (henceforth, dCDH) is uniquely suited to my setting of non-binary treatments. I thus use dCDH as my preferred method of pass-through estimation, while comparing its results to those of the more traditional TWFE. I present event study results as well as point estimates of cumulative average pass-through for different subsamples of gas stations, focusing primarily on income heterogeneity.

In the national event study, price differences between treatment and control group stations change very little over the seven months prior to a tax change, providing support for the parallel trends assumption. In the preferred dCDH specification, I find that national average pass-through of regional diesel tax changes into prices at the pump is approximately 92.4 percent in the year following a tax hike. The pass-through rate reaches its rough peak by the third month and, in a balanced sample, remains there for the two-year duration of my post-period. It is statistically significantly different from fully 100 percent, or "one-for-one", pass-through.

I then re-estimate event studies in subsamples defined by quintile of the municipality median income distribution, as well as in the group of gas stations in municipalities with population below the threshold (5,000) for inclusion in the income data. Prior price differences between treatment and control groups again show no evidence of non-parallel trends. Across a range of specifications, pass-through rates are at their lowest in the bottom income quintile and in the smallest towns lacking income data, while being higher and generally similar across the top four quintiles. The difference in average pass-through rate in the top quintile versus the bottom is in the range of 10-11 percentage points; in my preferred specification for inference, pass-through averages 84.8 percent in the bottom quintile and 95.4 percent in the top, and the difference in rates is statistically significant at the one percent level. The qualitative result of rising pass-through in income is robust to choices about the income measure (median versus per capita income; alternative quantiles), whether to weight observations (in the preferred specification, I weight by station-year sales volume), as well as the cross-sectional unit of observation (station, municipality, or region by income quintile).

I use these results and other suggestive analyses to discuss three extensions: implications for distributional welfare assessment; potential causal explanations; and non-tax fuel cost pass-through. I show that proportional fuel consumption (per dollar expenditure) drops in household expenditure decile, making fuel taxes broadly regressive; progressive pass-through (that is, rising in wealth) of the magnitude I find here offsets some of that regressivity. I also provide evidence that station branding patterns and (horizontal and vertical) market structure, despite being strongly predictive of pass-through rates in some cases, do not fully explain the observed income/pass-through relationship. Lastly, I show that pass-through is greater in response to crude oil price changes than to tax changes and varies little across income groups, which is consistent with a literature that finds larger consumer demand response to the latter than to the former (Davis and Kilian, 2011; Li et al., 2014).

This research contributes to a long economics literature related to pass-through (see, e.g., Jenkin, 1872; Weyl and Fabinger, 2013) that spans traditional sub-fields such as public finance (Poterba, 1996), industrial organization (Bonnet et al., 2013), and international economics (Cavallo et al., 2021).

Most specifically, my findings relate to the literature concerned with estimating the pass-through of energy taxes (Marion and Muehlegger, 2011; Fabra and Reguant, 2014), subsidies (Lade and Bushnell, 2022; Muehlegger and Rapson, 2022), and input costs (Borenstein et al., 1997; Ganapati et al., 2020; Muehlegger and Sweeney, 2022). In empirically documenting variation in fuel tax pass-through with income, I am highlighting a largely-ignored dimension of potential heterogeneity that has implications for the fairness of policy. In applying the de Chaisemartin and D'Haultfoeuille (2024a) method of DD estimation, I provide an early example of new techniques for improved panel estimation of treatment effects (alongside, e.g., Rico-Straffon et al., 2023) that are well-suited to pass-through estimation.

My findings also relate to the literature devoted to distributional welfare impacts of policy. Studies which focus on how progressive or regressive a cost change is tend not to allow for heterogeneous markup adjustment by firms, instead choosing a single pass-through rate to apply throughout the analysis. This practice is prevalent in the energy tax literature (Metcalf, 1999; West, 2004; Bento et al., 2009; Mathur and Morris, 2012) but is also employed in studies of the U.S. sales tax (Caspersen and Metcalf, 1994) and cigarette taxes (Gruber and Koszegi, 2004). My results suggest that assessment of pass-through heterogeneity across the wealth spectrum may be a worthwhile exercise before proceeding with a "uniform pass-through" assumption in welfare analysis.

1 Pass-through and welfare

Pass-through is "the diffusion throughout the community of economic changes which primarily affect some particular branch of production or consumption" (Marshall, 1890). Those changes can be costs, such as input costs or taxes, physically imposed on one part of a supply chain and passed through to others; and they can similarly be benefits, such as government subsidies for specific retail investments. Pass-through has broad value across economics because it is both an outcome of interest in its own right and a sort of "sufficient statistic" (Chetty, 2009) for answering questions about market structure and policy impacts (Jaffe and Weyl, 2013; Atkin and Donaldson, 2015; Pless and van Benthem, 2019).

In perfect competition, pass-through is entirely a function of elasticities of supply and demand. Equation 1 provides the mathematical definition:

$$\frac{dp_c}{dc} = \frac{\epsilon_S}{\epsilon_S - \epsilon_D} = \frac{1}{1 - \frac{\epsilon_D}{\epsilon_S}} \tag{1}$$

Pass-through of cost c to retail price p_c rises in the supply elasticity (ϵ_S) and falls in the absolute demand elasticity (ϵ_D). In the polar cases of either perfectly elastic supply ($\epsilon_S \to +\infty$) or perfectly *in*elastic demand ($\epsilon_D \to 0$), pass-through is identically 100 percent.

In *im*perfect competition, pass-through varies with not just the first derivative (elasticity) but also the second (convexity). Consider the formula for pass-through in monopoly with constant marginal costs c:

$$\frac{dp_m}{dc} = \frac{\frac{\partial p(q_m)}{\partial q_m}}{2\frac{\partial p(q_m)}{\partial q_m} + q_m \frac{\partial^2 p(q_m)}{\partial q_m^2}} \tag{2}$$

Equation 2 shows that monopoly pass-through can, in principle, be higher or lower than perfectly competitive pass-through – it depends on the demand convexity parameter $\frac{\partial^2 p(q_m)}{\partial q_m^2}$ (Seade 1985).[1] Under perfect competition, constant marginal cost (i.e., perfectly elastic supply) guarantees fully 100 percent pass-through. Under monopoly with linear demand, $\frac{\partial^2 p(q_m)}{\partial q_m^2} = 0$ and pass-through simplifies to a constant 50 percent. If, however, $\frac{\partial^2 p(q_m)}{\partial q_m^2} > 0$, then market power increases pass-through relative to that of perfect competition. If $\frac{\partial^2 p(q_m)}{\partial q_m^2}$ is positive and sufficiently large, then pass-through can exceed 100 percent.

While the relationship between pass-through and market power has received much attention, the relationship between pass-through and wealth has not. Demand elasticities are a fundamental determinant of pass-through, and it is plausible that they themselves vary with wealth. In retail auto fuel markets, for instance, it could be that demand is more inelastic in poorer areas because the poor have more fundamental, non-discretionary uses of auto fuel; or it could be that the *wealthy* have more inelastic demand because of a wealth-induced price insensitivity. Empirically, retail auto fuel (absolute) demand elasticities have been found to both rise with income (Hughes et al., 2008; Kayser, 2000) and fall with income (Kilian and Zhou, 2024; Houde, 2012). Ultimately, pass-through may vary with wealth not only through variable demand elasticities but also through correlations of wealth with other determinants of pass-through, such as supply elasticities (Marion and Muehlegger, 2011), vertical market structure (Bajo-Buenestado and Borrella-Mas, 2022), or tax salience (Kroft et al., 2024).

A positive cost shock – such as a tax or input cost increase – elicits a direct change in consumer surplus through two channels: (a) the additional cost of consumption maintained in the face of rising prices; and (b) the utility lost from reduced consumption. Pass-through physically measures the former (per unit consumption), which is the first-order approximation to the consumer surplus impact of a marginal tax change. Total individual welfare is also determined at least by (a) ownership of supply-side capital; (b) externalities; (c) other goods' prices and quantities that are affected in general equilibrium; and (d) the use of government revenues obtained through taxation. In this paper, however, I focus only on the utility derived directly from the purchase and consumption of energy. Sterner (2012) provides a fuller discussion of the various channels through which a tax affects welfare in the context of fuel markets.

2 Empirical setting and data

The Spanish retail automotive fuel market is oligopolistic and vertically connected to the upstream oil refining market.[2] Three companies (Repsol, Cepsa, and BP) own the nine oil refineries producing automotive fuel in Spain.[3] In my sample window, these companies owned a significant (29.5 percent) share of the national Hydrocarbon Logistics Company (CLH), the regulated monopoly responsible

[1]Ritz (2024) shows that *cost* convexity can also produce a positive relationship between market power and pass-through, all else equal.

[2]For background on the evolution of Spain's oil markets, see Contín-Pilart et al. (2009) and Perdiguero and Borrell (2007).

[3]Spain both imports and exports refined petroleum products. In 2011, net imports accounted for 15 percent of refined diesel, but imports have been deemed ineffective at exerting competitive pressure on the three domestic refining companies (National Commission on Competition, 2012).

for transportation and storage of oil products in Spain (National Commission on Competition, 2012). Most importantly for the present study, Spain's refiners are very active in the retail market. In my main analysis sample, 64 percent of retail gas stations offer diesel bearing the brand of a refiner, and 28 percent of those are fully vertically integrated. These companies have faced significant scrutiny from government and popular media alike, on the grounds of alleged collusion and some of the highest estimated retail margins nationally in all of Europe (see, for example, Noceda and Jimenez, 2015).

2.1 Price and other station attributes

One result of such scrutiny has been closer monitoring of pricing by gas stations. A mandate which went into effect in January 2007 requires all stations across the country (more than 12,000 today; National Commission on Markets and Competition, 2024) to send in their fuel prices to the Spanish government whenever they change, and weekly regardless of any changes. These prices are then posted to a web page - called the *Geoportal* - designed for consumer use.[4] From Spain's Ministry of Industry, Energy, and Tourism (MINETUR),[5] I obtain daily price data for retail diesel at 9,884 Spanish gas stations from January 2007 to June 2013 (diesel has an 80 percent share of all retail automotive fuel sales in Spain during this time period, according to data from the Spanish strategic petroleum reserve [CORES]). The price data come with cross-sectional attributes: location, brand, wholesale contract type, and amenities.[6] Figure 1 maps all Spanish gas stations, excepting those in the Canary Islands and the African-contintental territories of Ceuta and Melilla.

I add two further types of data to the station-level dataset recorded in Spain's Geoportal. First, I obtain station-year sales volumes from 2008 to 2014, also from MINETUR. Station-level sales volumes are useful because I am primarily interested in estimating population-average pass-through rates, not just unweighted averages across stations.[7] The merge of price and quantity data is imperfect, resulting in 89 percent of stations in the price data being successfully matched to annual sales at least once.

Second, I calculate two station-specific proxies for market power. One of these is a count of open stations within a 10-minute driving distance radius, weighted by $\frac{1}{1+d}$, where d is the driving distance (in minutes) between a pair of stations as measured by Google Maps. This count captures degree of spatial isolation from competitors.[8] The second is the proportion of stations within a 10-minute

[4]<https://geoportalgasolineras.es/geoportal-instalaciones/Inicio>; there is now a phone application as well. Appendix Figure A1 presents a screenshot of the *Geoportal* from 2015.

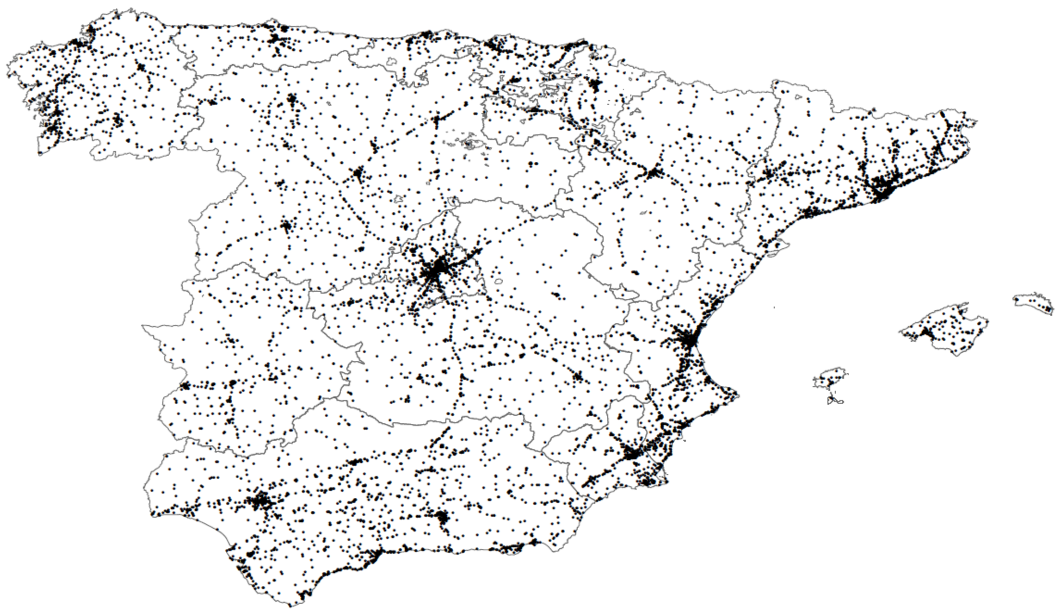
[5]This ministry has been renamed and reorganized multiple times over the past twenty years. At the onset of the mandate, it was called the Ministry of Industry, Tourism, and Trade. In 2011, it was renamed the Ministry of Industry, Energy, and Tourism. In 2016, it was renamed again to the Ministry of Energy, Tourism, and Digital Agenda. Currently, the *Geoportal* is managed by the Ministry for the Ecological Transition and Demographic Challenge.

[6]Station-specific *changes* in brand and contract type over time are thus unobservable. Spain's National Commission on Competition (2012) reports that "the relative positions of the three majors (Repsol, Cepsa, and BP) by number of service stations has remained more or less stable over time." Ownership changes at retail stations over time have overwhelmingly been exchanges among wholesale operators without refining capacity in Spain, and my parameterization of station brand into three groups for analysis – refiner brands, non-refiner wholesaler brands, and independent brands – is unaffected by such exchanges.

[7]While I would like to estimate demand elasticities, annual frequency of sales-volume observation is insufficient in my context of small, within-year tax changes.

[8]A station's relevant competitors are defined in part by typical travel patterns, including commuting (Houde, 2012). Lacking commuting data for the whole of Spain, I use the simpler, commonly used travel time based measure of competition (e.g.,

Figure 1: *Map of Spanish gas stations, 2007-2013*



Notes: The Canary Islands and Ceuta and Melilla territories are omitted from the map.

radius that share one's brand; this measure captures the degree of own-brand concentration in local markets. Both of these competition proxies vary cross-sectionally and over time due to entry and exit of stations.

2.2 Taxes

Fuel taxes are large as a proportion of total price in Spain – 45 percent, on average across station-days subject to the main tax of interest during my observation window. In most of Spain (excluding the Canary Islands, Ceuta, and Melilla) there are three taxes applicable to retail diesel: a national sales tax that rises twice, going from 16 to 18 and then 21 percent; a national excise tax ("Impuesto Especial de Hidrocarburos", or IEH) that changes once, from 27.8 to 30.7 Eurocents per liter (c/l); and a smaller excise tax ("Impuesto sobre las Ventas Minoristas de Determinados Hidrocarburos", or IVMDH) with both a national component (2.4 c/l) and an optional regional component (Spanish Tax Agency, 2024).[9] I rely on the regional component of this last tax for pass-through identification, because it is the only source of panel variation in tax levels. It is colloquially known as the *céntimo sanitario* (" health cent"), because of its modest size – 0 to 4.8 c/l – and the fact that its revenues were earmarked for public health expenditures (Court of Justice of the European Union, 2014).[10]

Doyle Jr. and Samphantharak, 2008; Miller et al., 2017). Genakos and Pagliero (2022) show, using fuel tax and price data from Greek Islands, that distance-restrictions on local retail fuel market definitions can lead to overestimation of pass-through when the number of nearby stations is small. In my context, however, the logic ends up working the other way, because I find pass-through rising in market power, rather than falling as in Genakos and Pagliero (2022).

[9]The sales tax is applied on top of the excise taxes; I remove the effect of the sales tax from the final retail price and use the resulting "pre sales tax" price as the outcome of interest.

[10]In 2013, the IVMDH was integrated into the IEH to comply with European Union Law. In 2014, the Court of Justice of the European Union (2014) ruled that, from 2002 through 2011, the tax was unconstitutional and its revenues must be returned.

At the first sub-national level of government, Spain has 17 autonomous communities; these are the "regions" to which I refer throughout the paper. Between 2008 and mid 2013, there are 14 discrete regional tax changes in 11 unique regions (three regions change their level twice), all of them increases.[11] Figure 2 documents the timeline of these tax changes, with average retail and crude oil (Brent benchmark) price trends provided in the background for context. The first tax hike occurs in July 2010, and each month thereafter has 0-4 additional, region-specific tax hikes. The increases are modest; the average size of the 14 tax changes is 2.97 c/L, which is about 2.6 percent of after-tax retail prices.

Figure 2: *Regional tax changes and prices over time*

Notes: The sample is all Spanish stations excluding the Canary Islands region and Ceuta and Melilla territories. Retail prices are monthly national averages, calculated from daily national averages weighted by station-level quantity sold. Crude oil prices are monthly averages of the Brent benchmark price. Tax changes are counts of regions experiencing a tax change in a given month.

The *céntimo sanitario* changes are not random; correspondence with the Spanish Ministry of Industry, Energy, and Tourism indicates that they are driven by regional government budgetary needs, specifically for health care as well as more generally. The tax changes are preceded in time by the Great Recession beginning in 2008, and also by a steady decentralization of health care provision responsibilities from the national government to regional ones (Costa-Font and Rico, 2006). My difference-in-differences identification strategy controls for national time-specific shocks to retail fuel prices as well as cross-sectional, station-specific determinants of fuel prices. To address the potential for endogeneity conditional on these controls, I examine treatment and control group trends in prices

[11] The regional tax changes apply identically to diesel and gasoline – the main alternative auto fuel to diesel – in eight of eleven regions. To the extent that substitution between diesel and gasoline use occurs, places where diesel and gasoline tax hikes deviate in magnitude may experience different diesel pass-through rates than places with identical gas tax hikes for the two fuels. To account for this possibility in analysis, I ultimately consider a robustness check which omits the three regions with such a deviation (Extremadura, Murcia, and Valencia).

predating the tax changes.

2.3 Income and population

I combine the station (price) and tax datasets with income data from Spain's Foundation for Applied Economic Studies (FEDEA). I collect per capita income and median income per declarant by municipality-year, from 2008-2013, for all municipalities with population greater than 5,000, in all but two of Spain's regions (data are not available from País Vasco and Navarra regions, which are not a part of Spain's Common Fiscal Regime Territory). I assign municipalities to quantiles; primarily, these are quintiles defined by municipalities' sample-wide average median income, but I also define deciles in the same way as well as quintiles by average per-capita income and 2010 median income.

Importantly, I retain the large subset of stations in municipalities too small to be included in the income data as a data point in the assessment of pass-through heterogeneity. While incomes in this more-rural subset are not measured, inspecting the relationship between income and population near the threshold of 5,000 is suggestive. Figure 3 allows for this inspection; I plot a binscatter of municipality median income and population obtained from Spain's National Statistical Institute. There is a clear positive relationship, with the appearance of concavity near the threshold, such that income is dropping fast in population nearest the region of interest (<5,000 population). It is certainly possible for the trend to break, but this figure suggests it is unlikely that the "no income" subgroup has incomes *higher* than the bottom income quintile.

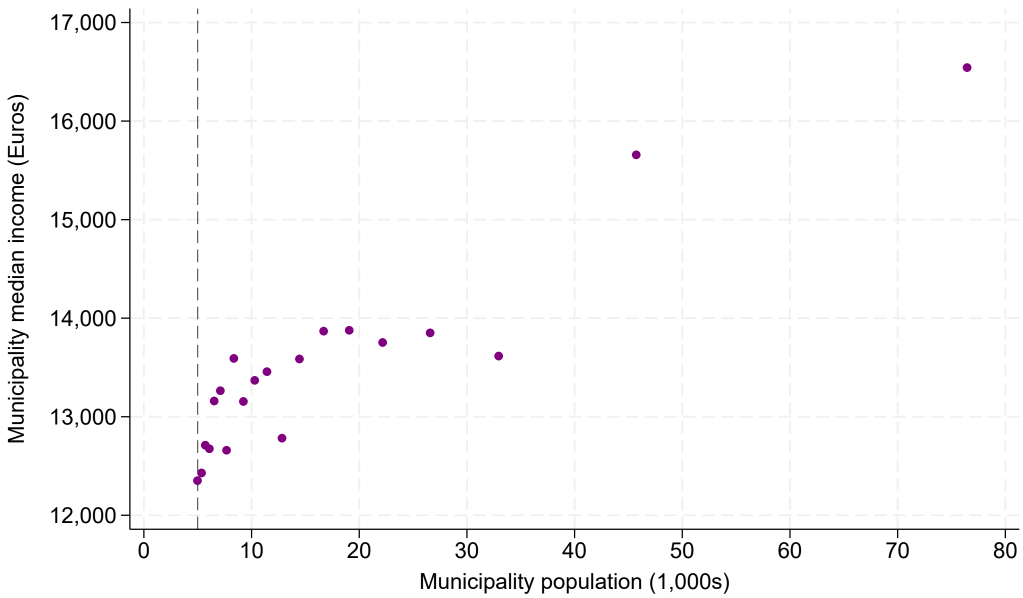
2.4 Main analysis sample

I start with 9,984 unique stations observed in the raw Geoportal data, spanning 17 regions and 2 territories. I drop stations in the one region (Canary Islands) and two territories (Ceuta and Melilla) that are not subject to the key IVMDH excise tax; these three areas each have their own separate value-added tax and are quite geographically isolated from other regions of Spain. This yields 9,187 stations across 16 regions. I drop a further 249 stations that are within a ten-minute drive of another station in a different region. Pass-through estimation at these "border" stations has a different interpretation than elsewhere, because a tax hike on one side of a cross-border market is an own-cost shock for the taxed station and a rival-cost shock for untaxed competitors – rather than a market-wide one (Stolper, 2016; Muehlegger and Sweeney, 2022).

For most of the analysis, I omit stations without sales volume data, since my primary interest is in estimating sales-weighted pass-through rates.[12] This restricts the analysis sample to 7,992 stations and a window of 2008-2013. I additionally omit stations in the País Vasco and Navarra regions, which are not covered by the income data. This leaves 7,564 stations available for estimation of the pass-through/income relationship. I summarize the attributes of these stations in Table 1. To construct the table, I first collapse station-year-month observations to station-level observations, weighting retail price and tax level by station-year diesel sales volume. I then calculate sample-wide

[12] Appendix Figure A2 shows how the income distribution across stations differs between the full sample and the sample without sales data.

Figure 3: *Population and median income across municipalities*



Notes: Plotted data points are obtained from a binned scatterplot of municipal average population and municipal median annual household income; observations are grouped into twenty equal-sized population bins, and then averages of both population and income are calculated within bins. The input data sample is truncated above at a population of 100,000, and it excludes the three regions and two territories for which income, population, or tax data are unavailable. The dashed line indicates a population of 5,000, the cutoff for being included in the income dataset.

means from these station-level observations without weights.[13] The table presents summary statistics nationally as well as in each of five income quintiles and the group of most-rural stations for which income data are unavailable.

The national average retail price (excluding sales tax, unweighted by station sales volume) is 100.43 c/l, and all subgroups have an average price within 0.5 cents of that. 64 percent of stations nationally are "refiner-branded", which is to say, have a contract for branded fuel from a Spanish oil refiner. 14 percent have a contract for branded fuel with a wholesaler only ("wholesaler-branded"), and the remaining 22 percent do not have contracts for branded fuel ("independent-branded"). 24 percent of stations are fully vertically integrated with a refiner or wholesaler. Meanwhile, several attributes generally rise in income quintile. Relatively higher-income areas tend to have higher sales, greater concentrations of refiner and wholesaler brands, fewer independent-branded stations, less vertical integration, fewer nearby rivals, and higher population and population density.

[13]Appendix Table A1 presents the analogous statistics *with* such weights. Appendix Table A2 provides summary statistics for some additional station attributes.

Table 1: *Summary statistics*

	All	No Inc	Q1	Q2	Q3	Q4	Q5
Retail diesel price (c/L)	100.43 (4.21)	100.06 (4.65)	100.19 (4.22)	100.42 (4.10)	100.61 (3.94)	100.45 (4.03)	100.94 (3.82)
Regional retail tax (c/L)	1.94 (0.97)	1.83 (1.13)	1.81 (0.80)	1.93 (0.81)	1.94 (0.88)	1.91 (0.88)	2.16 (0.92)
Mean annual sales (unweighted L)	2,023.98 (2,263.83)	1,607.29 (2,997.35)	1,571.12 (1,023.31)	1,861.20 (1,231.05)	2,146.07 (2,178.65)	2,348.46 (2,051.68)	2,536.30 (1,908.52)
Refiner-branded (0/1)	0.64 (0.48)	0.62 (0.49)	0.62 (0.48)	0.63 (0.48)	0.66 (0.47)	0.66 (0.47)	0.67 (0.47)
Wholesaler-branded (0/1)	0.14 (0.34)	0.10 (0.30)	0.12 (0.32)	0.13 (0.34)	0.14 (0.34)	0.16 (0.36)	0.19 (0.39)
Independent-branded (0/1)	0.22 (0.41)	0.28 (0.45)	0.26 (0.44)	0.24 (0.42)	0.20 (0.40)	0.18 (0.39)	0.15 (0.36)
Brand vertically integrated (0/1)	0.24 (0.43)	0.15 (0.36)	0.13 (0.34)	0.21 (0.41)	0.28 (0.45)	0.31 (0.46)	0.35 (0.48)
# rivals, distance-weighted	1.24 (1.28)	0.43 (0.59)	0.67 (0.69)	1.03 (0.99)	1.36 (1.22)	1.79 (1.28)	2.17 (1.46)
Proportion of rivals sharing one's brand	0.30 (0.34)	0.44 (0.43)	0.24 (0.34)	0.23 (0.29)	0.25 (0.30)	0.24 (0.25)	0.24 (0.23)
Municipality population (1,000s)	154.18 (508.37)	1.96 (1.34)	16.81 (17.13)	33.42 (44.73)	31.04 (28.53)	144.57 (155.44)	555.97 (984.54)
Municipality pop. density (1,000s per km2)	1.10 (2.53)	0.06 (0.14)	0.16 (0.23)	0.32 (0.78)	0.83 (1.67)	1.56 (3.15)	3.09 (3.65)
Mean of median income across years	14,604.02 (3,510.71)	. (.)	9,352.88 (885.20)	11,353.93 (524.67)	13,208.20 (510.51)	15,448.52 (644.41)	18,456.29 (2,547.42)
Observations	7,564	2,245	634	896	816	1,354	1,619

Notes: Unweighted means and standard deviations (in parentheses) are calculated from station-level observations. "All" refers to all stations in the main analysis sample. "No Inc" indicates stations in municipalities too small to have income data reported. *Q*1 through *Q*5 denote quintiles of the municipality median income distribution. Brand vertically integrated stations are owned and operated by a wholesaler or refiner. Proportion of rivals sharing one's brand is calculated within a ten-minute radius of each station and takes a value of one in the case of zero rivals. # of rivals is similarly calculated within a ten-minute radius and is a weighted sum, with weights equal to inverse travel time in minutes to each rival.

3 Methods

3.1 Difference in differences with staggered treatment

Difference-in-differences – comparing "treated" units to untreated ones, before versus after the treatment is applied – is a common method of causal estimation across the social sciences (e.g., Roth, 2022; Wing et al., 2024). In the frequent case that the treatment of interest is staggered in time across many units being studied, two way fixed effects (TWFE) regression is a common strategy for specifying the DD comparison. The TWFE model is as follows in my context, which involves a non-binary treatment:

$$P_{it} = \alpha + \beta \text{Tax}_{it} + \theta_t + \phi_i + \epsilon_{it}. \tag{3}$$

P_{it} is the retail price of auto fuel and Tax_{it} is the region-specific tax level, both at station i in year-month t. α is a constant, θ_i and ϕ_t are station and year-month fixed effects, respectively, and ϵ_{it} is a mean-zero error term conditional on the predictors. The coefficient β is interpreted as the average pass-through rate over the full post-period after a tax change. Analogously, one might estimate an event study model, where Tax_{it} is replaced by a vector of binary variables (D_{it}^{j}) equaling one if an observation is $j \in [a, b]$ months from a tax change in its region, and relative-month specific coefficients β_j are estimated instead of β:

$$P_{it} = \alpha + \Sigma_{j=a}^{b} \beta^{j} D_{it}^{j} + \theta_t + \phi_i + \epsilon_{it}. \tag{4}$$

Identification of average treatment effects via DD requires that two key assumptions hold; first, that a group's current outcome does not depend on its future treatments ("no anticipation"); and second, that in the absence of treatment, outcomes among treated and untreated groups would have evolved in parallel ("parallel trends") (Roth, 2022).[14] The credibility of these assumptions can be examined visually with an event study plot showing pre-treatment data points. There is no uniformly applied statistical criterion for validating the parallel trends assumption, but the individual statistical significance of the pre-treatment coefficients is common. Roth (2022), however, shows how typical statistical tests of the parallel trends assumption tend to be under-powered.

While non-parallel trends (in the treated group versus the control, predating treatment) are widely considered the main threat to credible estimation of treatment effects in the DD setup, a new and rapidly evolving literature has highlighted another significant threat to DD identification using TWFE: heterogeneous treatment effects. Consider that, in the case of binary and staggered treatment, the TWFE estimator is mathematically a weighted average of DD treatment effects across all pairs of units with one unit switching treatment (and the other not), for every observed pre-versus-post time period comparison (Goodman-Bacon, 2021). This includes, for example, comparing a group A that switches from untreated in period 1 to treated in period 2 and a group B that is treated in both periods. If treatment effects are constant, then taking the pre-post difference in group B removes the effect of group B's treatment and makes this DD comparison valid. But if treatment effects evolve over time, then group B's period specific treatment effects do not cancel each other out in the pre-post difference, and one treatment effect enters the TWFE average treatment effect calculation with negative weight. These negative weights result in the TWFE estimator not being a convex combination of the individual DD comparisons and not having the "no sign reversal" property (that is, it is possible for TWFE to produce an estimate of the wrong sign relative to the true value; de Chaisemartin and D'Haultfoeuille, 2023).

A second problematic type of comparison is made when the treatment is additionally non-binary (as is common in published papers using DD; de Chaisemartin and D'Haultfoeuille, 2023): a group A as above, becoming newly treated in period 2, and a group B that is also treated in period 2 but with a smaller treatment *size*. This comparison's validity requires constant treatment effects not just over time, as above, but also across groups. If the latter is not the case, then comparing a more-treated group to a less-treated one conflates treatment size with treatment effect.

Several recent papers – including Callaway and Sant'Anna (2021); Sun and Abraham (2021);

[14] The DD also requires the stable unit treatment value assumption (SUTVA), which rules out spillover and general equilibrium effects (Roth, 2022). In my context, one station's treatment does indeed affect other, competing stations' outcomes, but these competing stations are not in the control group, because treatment assignment occurs at the region-level. The exception is with cross (regional) border competition, which is why I omit stations near cross-border rivals from the main analysis sample.

Borusyak et al. (2024); de Chaisemartin and D'Haultfoeuille (2024a) – have proposed (and provided code for) alternative DD estimators that are robust to heterogeneous treatment effects under certain conditions, by virtue of avoiding the "forbidden comparisons" described above. I rely here on the methods of de Chaisemartin and D'Haultfoeuille (2024a) to estimate sample-wide and heterogeneous (sub-sample) treatment effects. The treatment in question – a fuel tax increase – is staggered and non-binary; de Chaisemartin and D'Haultfoeuille (2024a) are, to date, the only researchers to propose an estimator for a non-binary treatment. Their estimator, which I refer to as dCDH (computed in Stata with the -did_multiplegt_dyn- package), compares treated groups with as-yet-untreated groups, and it allows for dynamic (that is, lagged) treatment effects.

3.2 Implementation

Following de Chaisemartin and D'Haultfoeuille (2024a), let $Y_{g,t}$ be the retail fuel price at station g in year-month t. F_g is the month in which station g's tax changes. Station g's comparison group is g', which consists of all other stations having (a) the same initial tax level and (b) not yet been treated as of month F_g. A DD estimate of station g's treatment effect in post-treatment month ℓ is:

$$\mathrm{DD}_{g,\ell} = \left(Y_{g,F_g-1+\ell} - Y_{g,F_g-1}\right) - \sum_{g':F_{g',1}=F_{g,1},F_{g'}>F_g-1+\ell} \left(\frac{W_{g',F_g-1+\ell}}{N^g_{F_g-1+\ell}}\right)\left(Y_{g',F_{g'}-1+\ell} - Y_{g',F_{g'}-1}\right) \quad (5)$$

The baseline, or pre-, period is fixed here to the month immediately preceding station g's tax change; other pre-periods are reserved for "placebo tests" of the parallel trends assumption. $N^g_{F_g-1+\ell}$ is the number of observed stations in g' in a given post-treatment month, and $W_{g',F_g-1+\ell}$ is a weight equal to annual quantity sold at a given station in g' and corresponding to month $F_g - 1 + \ell$ (de Chaisemartin et al., 2024). Thus, control group pre-post differences are aggregated into a weighted average before being subtracted from the treatment-station pre-post difference. Month-specific event study effects DD_ℓ can then be computed as the average of $DD_{g,\ell}$ across all stations g,

$$\mathrm{DD}_\ell = \sum_{g:F_g-1+\ell\leq T_g} \frac{W_{g,F_g-1+\ell}}{N_\ell} DD_{g,\ell}, \quad (6)$$

where the weight W used is now quantity sold by *treated* station g. Equation 6 is unbiased conditional on the parallel trends and no-anticipation assumptions (de Chaisemartin and D'Haultfoeuille, 2024a).

The -did_multiplegt_dyn- command in Stata implements the estimation of Equations 5 and 6. It also estimates an average total effect per unit of treatment, which conveniently translates exactly into an average pass-through rate in my context. Let T_g be the last observed month for which there is an as-yet-untreated station remaining. Furthermore, define $D_{g,\ell}$ as the non-binary, discrete size of the tax faced by station g in post-month ℓ. Then, pass-through $\rho = \frac{\Delta P}{\Delta t}$ can be estimated with

$$\hat{\rho} = \frac{\sum_{\delta=1}^{T_g-F_g+1} w_\ell \mathrm{DD}_\ell}{\sum_{\delta=1}^{T_g-F_g+1} w_\ell \sum_{g:F_g-1+\ell\leq T_g} \frac{W_{g,F_g-1+\ell}}{N_\ell}\left(D_{g,F_g-1+\ell} - D_{g,1}\right)}, \quad (7)$$

where weight w_ℓ equals the proportion of all observed treated station-months that fall in month ℓ.

I estimate Equations 5-7 primarily in the full main analysis sample of stations and in each of six income groups separately – five quintiles and the rural subsample that is missing income data. I make several parameterization choices that define my base specification. First, I use quintiles defined by the municipality median income distribution, because rightward skew in household incomes causes a *mean* income measure to produce less representative quintile assignments. Second, I estimate policy effects for each month $\ell \in [1, 12]$ after a tax change, as well as six pre-treatment placebo effects for $\ell \in [-6, -1]$. Third, I include station-annual sales volume weights, to obtain an average pass-through rate across all units of diesel sold. Fourth, I cluster standard errors by province; while treatment is determined by region, which is one administrative level above province, there are only 17 regions in Spain – 14 of which get used in the main, income-focused analysis – whereas there are 55 provinces. Using the latter as cluster level avoids the "too-few clusters" problem that leads to downward bias in standard errors (Cameron and Gilbert, 2015). I test the robustness of my results to alternative choices in all four of the enumerated areas, as well as to alternative sample definitions and observation level.

4 Results

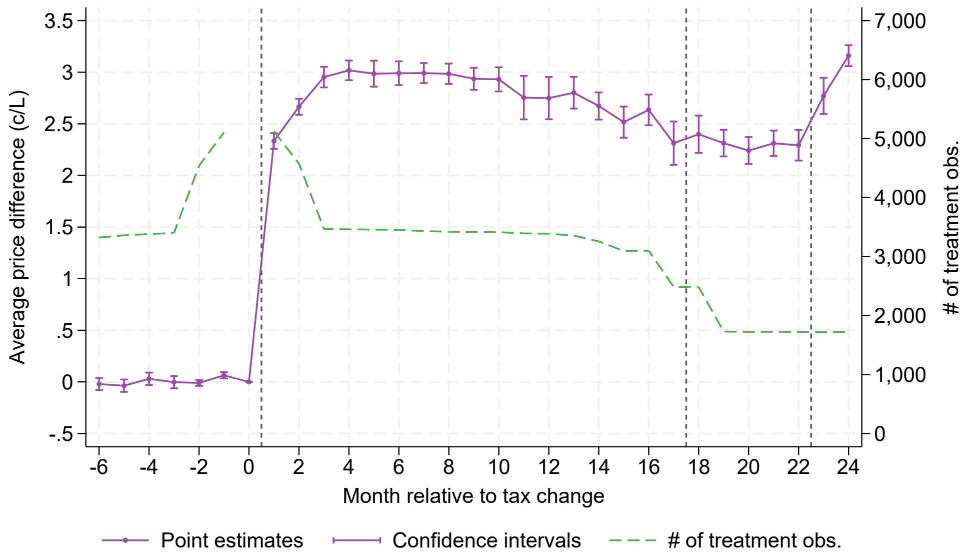
4.1 Overall fuel tax pass-through

The event studies in Figure 4 plot estimated differences in prices between the treatment and control groups as well as the number of treatment stations, for $\ell \in [-6, 24]$, in the full main analysis sample spanning fourteen regions of Spain. The point estimates are obtained via Equations 5 and 6. Since Equation 5 uses month 0 (that is, the month before a station's tax changes) as the pre-treatment data point in the DD design, the treatment/control price difference in month 0 is normalized to zero. Panel B differs from Panel A only through restricting the treatment group to stations for which effects are estimable for the entire duration of the observation window. Thus, Panel B serves to facilitate assessment of compositional effects over time in the "unrestricted" main analysis sample of Panel A.

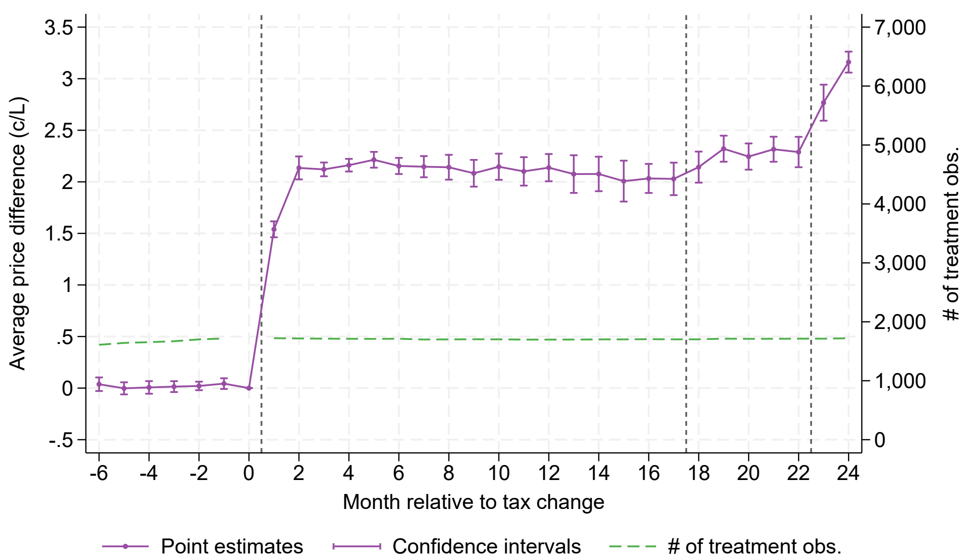
The levels and trend of the estimates for months -6 through -1 provide evidence on the parallel trends assumption. The pre-treatment trends in Figure 4 appear roughly flat. In Panel A, one of the six pre-treatment placebo effects ($\ell = -1$) is significantly different from zero at the five percent level. The average placebo effect is 0.003 and is not significantly different from zero (p=0.844). A straight line fitted by OLS through point estimates for periods -6 to 0 is also insignificant, with a slope of 0.008 and a p-value of 0.24. Put together, the evidence does not suggest a violation of non-parallel trends in the overall station sample.

Estimates jump significantly in the month of the tax change and remain elevated for the duration of the 24-month analysis window. The month-1 estimate is the lowest in this window; while this may indicate a delay in the full pass-through response, it is also a mechanical result of tax changes happening at various points within a month, such that the average tax level in that month is in *between* the pre- and post-treatment tax levels. After month 1, the post-treatment price impact in Panel A hovers between 2.67 and 2.96 cents per liter (c/l) for the rest of the first year, as compared to an average tax change of 2.98. The average price impact begins to drop in month eleven, levels out in month eighteen, and rises again in month 23. Notably, these trend breaks coincide with decreases in treatment N. Panel B, meanwhile, shows a comparatively flat trend all the way to month

Figure 4: *Full-sample average price impacts over time (24 months)*



(a) *Average price impacts, full national sample*



(b) *Average price impacts, subsample with no compositional change*

Notes: Point estimates and confidence intervals are for monthly DD coefficients (DD_ℓ) estimated using Equations 5 and 6, where $\ell \in [-6, 12]$ and month zero is omitted. # of treatment observations, measured on the right y-axis, indicates the number of stations used in estimation of each period's effect. Panel A uses the main analysis sample, while Panel B uses the subsample of stations for whom all post-treatment monthly effects can be estimated. Observations are station-year-months and are weighted by station-year diesel quantity sold. Standard errors are clustered by province. Vertical dashed lines denote, from left to right, the first tax change for stations in all treated regions, a second tax change for stations in Extremadura, and a second tax change in Murcia.

17 when using a balanced-panel treatment group. Two changes in the trend are apparent in months 18 and 23, which coincide perfectly with second tax changes in Extremadura and Murcia regions. The balanced-panel restriction reduces treatment N significantly (from a peak of about 5,000 in the unrestricted sample to about 2,000). Based on these results, I use twelve months as my primary post-period window, to avoid compositional effects and retain a larger monthly treatment N.

Table 2 presents points estimates of and standard errors for the cumulative average pass-through rate nationally at 6, 12, and 18 months out, according to Equation 7. Average pass-through is 91-93 percent in the preferred, weighted dCDH specifications of Panel A and 88-91 percent in the unweighted specifications of Panel B (columns 1-3). These estimates are in all cases significantly different from 100 percent, or fully one-for-one, pass-through. 12- and 18-month cumulative pass-through estimates are very similar (columns 2 and 3). TWFE yields slightly larger point estimates than dCDH, by 2.5 and 4.4 percentage points in Panels A and B, respectively (column 4 minus column 2). Appendix Table A3 presents analogous results with the main analysis sample *including* País Vasco and Navarro, for which income data are unavailable; the results are qualitatively similar.

Table 2: *Full-sample average tax pass-through*

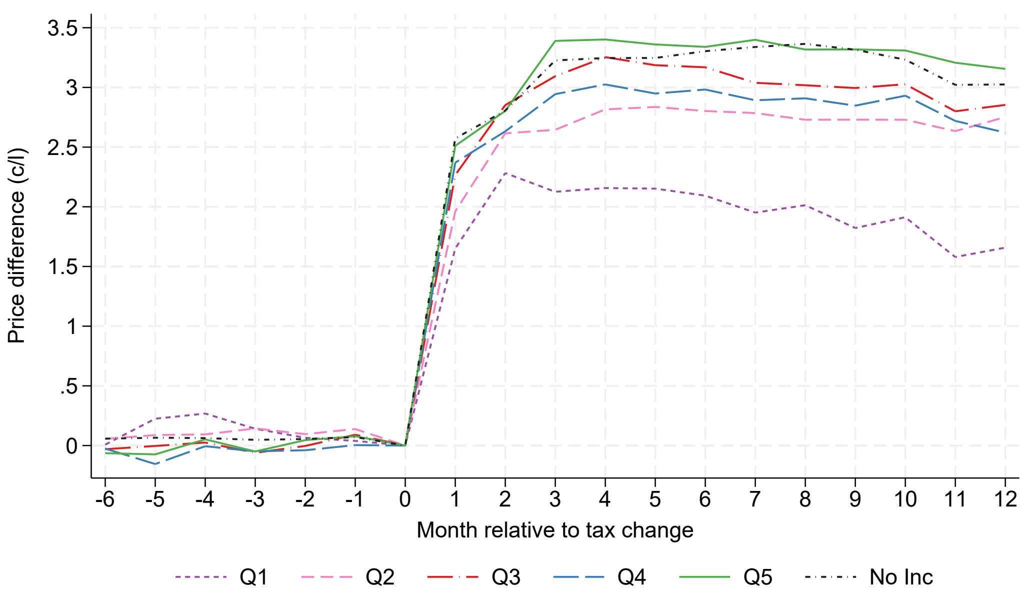
	dCDH			TWFE
	(1)	(2)	(3)	(4)
Panel A. Sales-weighted				
Average pass-through	0.932	0.924	0.913	0.947
	(0.015)	(0.015)	(0.015)	(0.029)
N	23,538	44,001	61,778	426,161
Post-period	6	12	18	12
Panel B. Unweighted				
Average pass-through	0.881	0.905	0.901	0.949
	(0.006)	(0.007)	(0.010)	(0.026)
N	29,449	53,493	74,604	487,000
Post-period	6	12	18	12

Notes: Estimates are average pass-through rates of tax changes into retail prices (ρ; cent/liter price change per cent/liter tax change) obtained via Equations 5-7 in columns 1-3 ('dCDH') and via Equation 3 in column 4 ('TWFE'). An observation is a station-year-month. 'N' reports the number of treatment-group post-period observations used in columns 1-3 and the number of total observations used in column 4. The two Panels (A and B) show results with and without analytic weights equal to station-year diesel quantity sold, respectively. 'Post-period" gives the number of months included in estimation. Standard errors are clustered by province.

4.2 Fuel tax pass-through by income group

Figure 5 presents event studies identical to that of Figure 4 except now separately for each of six income groups: the five quintiles of municipality median income, and the set of stations whose towns are too small for inclusion in the income data. Because of the number of event studies depicted here, I omit confidence intervals from the figure for visual clarity; Appendix Figures A3-A8 present income group specific plots that include confidence intervals. While pre-period coefficients are sometimes statistically different from zero (10 out of 36 times across all income groups), a best fit line for pre-period coefficients is not significant in any of the six income groups (p-values range from 0.14 to 0.83).[15] Once treatment occurs, all income groups show a mean shift in prices. The size of the mean shift varies from roughly 2-3.5 c/l; however, the ordering of income groups vertically cannot be interpreted as reflective of relative pass-through rates, because the *sizes* of tax changes experienced are not uniform across income group. The absolute-price impacts shown in Figure 5 must be adjusted to account for tax-change magnitudes.

Figure 5: *Income and absolute price impacts of tax changes*



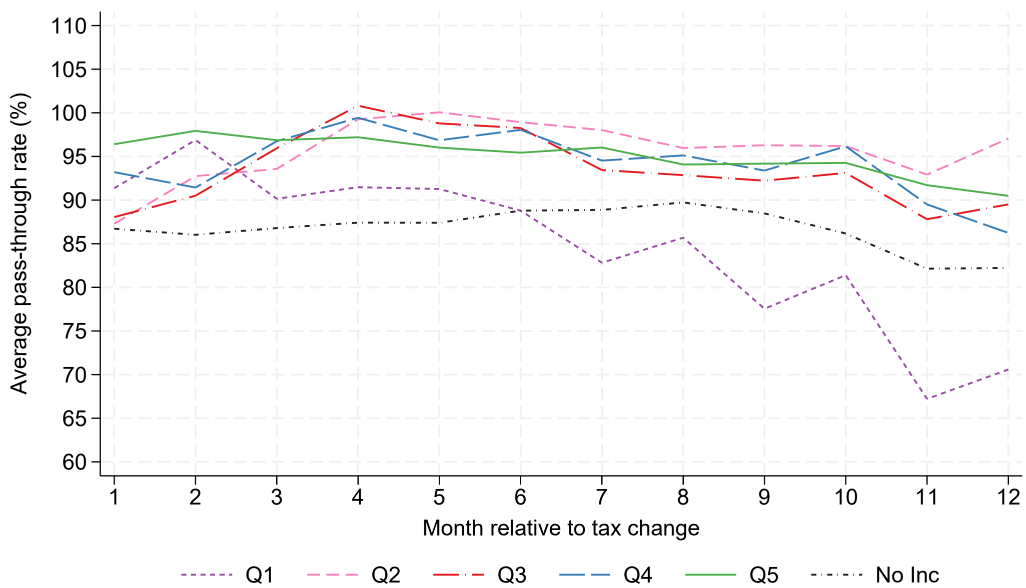
Notes: Lines are constructed from monthly DD point estimates of average treatment effect (DD_ℓ) by income group, estimated using Equations 5 and 6, where $\ell \in [-6, 12]$ and month zero is omitted. Observations are station-year-months and are weighted by station-year diesel quantity sold. Standard errors are clustered by province. Q1-Q5 indicate quintiles of the municipality median income distribution; "No Inc" denotes the group of stations in towns too small to be included in the data.

Figure 6 does exactly that. I calculate month-specific pass-through rates analogously to Equation

[15] I inspect the bottom income quintile group further, since it is the only group for which visual inspection suggests the possibility of a qualitatively meaningful pretrend. The pre-period best fit line has a slope equal to -0.023, and the p-value for a significant test of this slope is 0.335. Roth (2022) shows that pretrends tests can be underpowered, but I find (using Roth's diagnostic calculation) the slope test to be moderately powered in my context: if the true slope were -0.023, the power of my significance test would be 66 percent. Moreover, removing the effect of a pretrend with slope -0.023 from the post-period coefficients does not meaningfully affect the ordering by magnitude of income group specific pass-through rates.

7 (and following de Chaisemartin et al., 2024), dividing each monthly estimated price impact by a weighted average of tax-change sizes in that period. Higher income quintiles experience larger absolute price impacts in part because they experienced larger tax changes, but there remains evidence of pass-through heterogeneity: the range of income group specific rates varies between 10 and 27 percentage points across months. Starting in month 3, the lowest income quintile (Q1) and the no-data group (which, I argue, likely has even lower incomes than Q1) have consistently lower pass-through estimates than the other four quintiles.

Figure 6: *Income and pass-through, all groups*



Notes: Lines are constructed from monthly DD point estimates of average pass-through rate (ρ; cent/liter price change per cent/liter tax change) by income group, estimated using Equations 5 and 6 and a monthly analog of Equation 7, where $\ell \in [1, 12]$. Observations are station-year-months and are weighted by station-year diesel quantity sold. Standard errors are clustered by province. Q1-Q5 indicate quintiles of the municipality median income distribution; "No Inc" denotes the group of stations in towns too small to be included in the data.

I present binary comparisons of income groups in Figure 7, to provide a clearer picture of how different pass-through rates are at the bottom and top of the income spectrum. Panel A compares Q1 to Q5, while Panel B compares the rural group without income data to Q5. In both panels, the lower-income group has an average pass-through rate that is consistently lower than that of Q5. Month-12 pass-through is lower than month-1 pass-through in all three groups, with the steepest downward slope over time apparent in Q1. Confidence intervals in Panel B shed some qualitative light on the significance of differences between group-specific pass-through rates, but because I cluster standard errors at the province level here rather than the station level, I cannot formally test for this without bootstrapping (de Chaisemartin and D'Haultfoeuille, 2024b). However, I *can* do so with a slightly different version of the analysis that uses municipality-month observations and municipality clustering of standard errors. Table 3, which presents twelve-month average pass-through rates by income group, includes results from this municipality-level setup.

I obtain estimates in Table 3 via Equations 5-7, using each income group separately. Panel A uses the primary, station-month observation level, while Panel B uses the municipality-month level that facilitates hypothesis testing about heterogeneity. The lowest pass-through estimate is that of Q1, at 84.8 and 85.4 percent among stations and municipalities, respectively. The next lowest is that of the rural, no-income-data group, at 86.8 in both cases. The remaining four income quintiles have pass-through rates between 93 and 96 percent. Following the instruction of de Chaisemartin and D'Haultfoeuille (2024b) and using the results of Panel B, I calculate the variance of the difference in two quintile-specific pass-through rates by summing the estimated variances of the two pass-through rates themselves. I am able to reject the null of equal pass-through rates in Q1 and all other quintiles; p-values are less than 0.01 in all cases. Significance tests of the difference in pass-through between the no-income group and each of Q2 through Q5 yield p-values in the range of 0.1-0.2.[16]

Table 3: *Average tax pass-through by income group*

	No Inc (1)	Q1 (2)	Q2 (3)	Q3 (4)	Q4 (5)	Q5 (6)
Panel A. Station-year-month level						
Average pass-through	0.868	0.848	0.958	0.935	0.943	0.954
	(0.051)	(0.013)	(0.019)	(0.011)	(0.022)	(0.017)
N	11,987	4,981	7,167	5,867	8,851	5,136
Panel B. Municipality-year-month level						
Average pass-through	0.868	0.854	0.937	0.938	0.950	0.951
	(0.050)	(0.015)	(0.022)	(0.017)	(0.020)	(0.009)
N	9,670	2,009	1,775	1,666	1,182	678

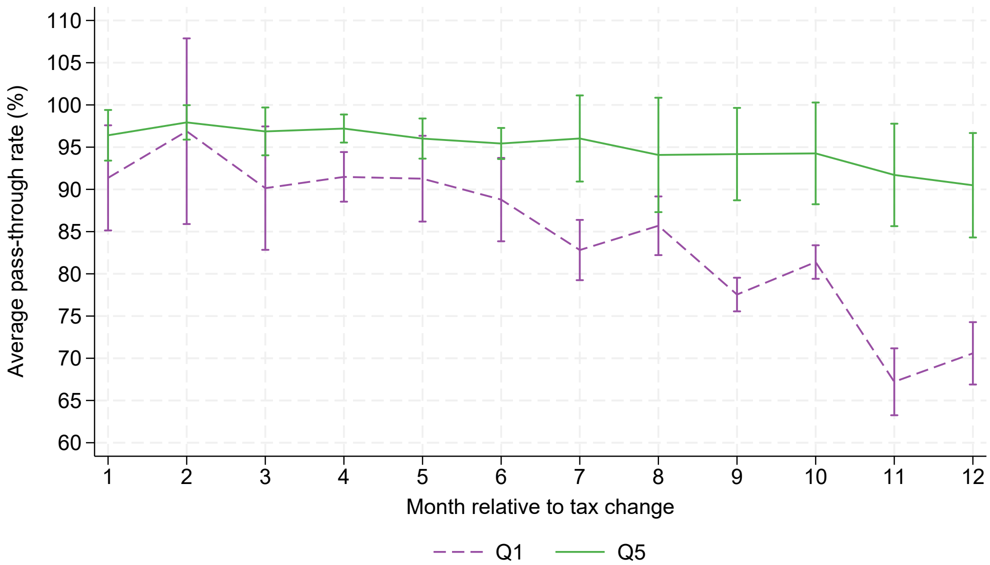
Notes: Estimates are average pass-through rates of tax changes into retail prices (ρ; cent/liter price change per cent/liter tax change) obtained via Equations 5-7, for each of six income groups. Panel A displays results with station-year-month observations; Panel B results are with municipality-year-month observations. The post-tax estimation window is 12 months. Observations are weighted by station- (or municipality-) year diesel quantity sold. Standard errors are clustered by province. 'N' reports the number of treatment-group post-period observations used. Q1-Q5 indicate quintiles of the municipality median income distribution; "No Inc" denotes the group of stations in towns too small to be included in the income data.

Table 4 summarizes results from other specifications of pass-through rate estimation by income group. Panel A restricts the dCDH analysis to using a subsample of stations for whom all twelve monthly post-treatment effects can be estimated; Panel B omits station sales weights from the analysis; Panel C uses per capita income instead of median income to define quintiles;[17] and Panel D uses two

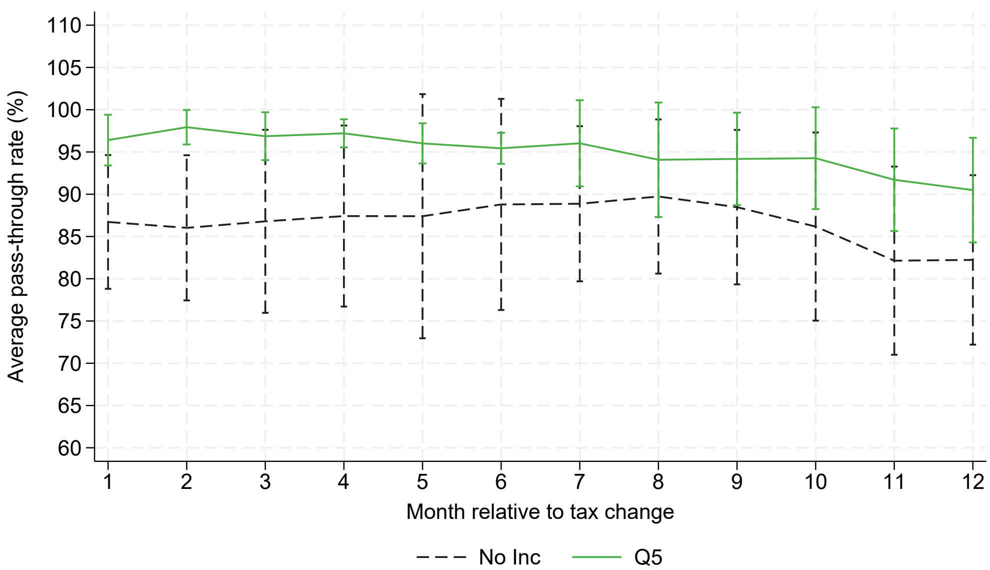
[16]Using an alternative observation level of region by income group by year-month – see Appendix Table A5 – pass-through rates in the no-income group additionally become significantly different from each of Q2/Q3/Q4/Q5 rates.

[17]Per capita income quintile assignments deviate from median income assignments because 55 of 226 municipalities assigned

Figure 7: *Income and pass-through, binary comparisons*



(a) *Avg. pass-through: income quintiles 1 and 5*



(b) *Avg. pass-through: missing income versus quintile 5*

Notes: Lines are constructed from monthly DD point estimates of average pass-through rate (ρ; cent/liter price change per cent/liter tax change) by income group, estimated using Equations 5 and 6 and a monthly analog of Equation 7, where $\ell \in [1, 12]$. Observations are station-year-months and are weighted by station-year diesel quantity sold. Standard errors are clustered by province. Panel A displays quintiles 1 and 5 of the municipality median income distribution, while Panel B displays the no-income group and quintile 5. Vertical lines are confidence intervals.

way fixed effects. The relative pattern across income groups is consistent across all of these dCDH specifications. Pass-through is uniformly smallest among the two lowest-income groups (the rural group and Q1) – mostly below 90 percent – and relatively higher in Q2 through Q5 – between 93 and 97 percent. Notably, TWFE yields larger estimated differences across groups than dCDH; in particular, Q3-Q5 reach 97-100 percent in the TWFE estimation. This implies that the "forbidden comparisons" included in TWFE bias estimates of (the magnitude of) heterogeneity upwards in this context.

Appendix Tables A4-A6 present further robustness checks on estimates by income quintile. Table A4 presents results from using region-level clustering of standard errors or region by income-quintile observations. Table A5 presents results from three alternative sample definitions: omitting the three regions whose diesel and gasoline tax hikes have different sizes; including stations within a ten-minute drive of a station in a different region; and excluding municipalities lacking a full set of annual income measures from 2008-2013. Finally, Table A6 presents results from defining income quintile according to 2010 values instead of 2008-2013 averages and from defining the sales weight according to station annual mean across years instead of the station total from the relevant single year. The pattern of results across income groups does not change significantly in any of these iterations of the analysis.

5 Extensions

5.1 Distributional welfare

A positive correlation between tax pass-through and income is noteworthy because of its potential implications for the distributional welfare of taxation. A typical distributional welfare calculation for a tax compares the percentage of a household's budget devoted to the taxed good at different quantiles of the household income or expenditure distribution.[18] This "normalized" fuel expenditure measure is a common way of accounting for declining marginal utility of consumption in wealth. I construct a similar measure, fuel consumption (quantity) per dollar total expenditure, using consumption and expenditure data from 2010 and 2013 iterations of the Spanish Household Budget Survey.[19] Figure 8 shows that relatively poorer households tend to consume relatively more fuel per dollar of total expenditure. From this figure, it appears that the *céntimo sanitario* is regressive.

The typical calculation described above is a simplification; in particular, it omits from consideration the effect of tax-induced pass-through on expenditure as well as the effect of tax-induced demand response on consumption. If pass-through is uniform, and if demand is perfectly inelastic, then these effects drop out of the distributional calculation. The point of my analysis, however, is to show that pass-through is non-uniform and, more importantly, correlated with income. Figure 9

to Q1 by per capita income (24 percent) are in a higher quintile when using median income, and these have a higher average pass-through rate than the remaining 171 municipalities assigned to Q1 using either variable. At the same time, there are also 55 municipalities assigned to Q1 by median income but a higher quintile by per capita; these exhibit relatively *lower* average pass-through.

[18] Poterba (1991) shows that household expenditure is a more accurate proxy for lifetime wealth than income.

[19] I use normalized fuel consumption *quantity* (as do Harju et al. (2022)) rather than expenditure, because it removes the effect of variable fuel prices across space. The plots look very similar using the expenditure variable (results available upon request).

Table 4: *Average tax pass-through by income group, alternative specifications*

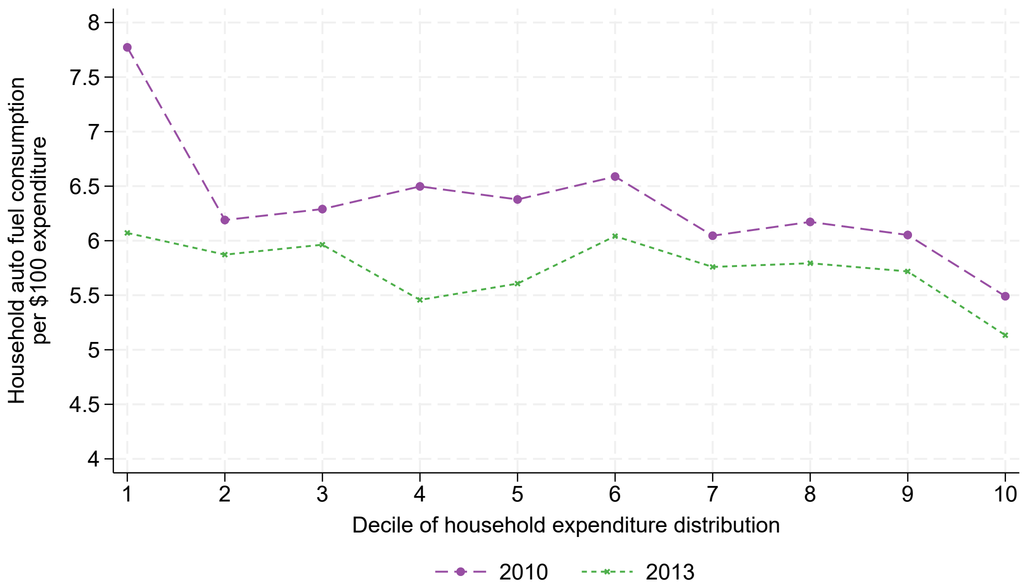
	No Inc (1)	Q1 (2)	Q2 (3)	Q3 (4)	Q4 (5)	Q5 (6)
Panel A. Subsample with no compositional change						
Average pass-through	0.865	0.847	0.966	0.935	0.948	0.947
	(0.055)	(0.013)	(0.019)	(0.012)	(0.023)	(0.020)
N	10,817	4,878	6,968	5,628	8,214	4,024
Panel B. Unweighted						
Average pass-through	0.889	0.857	0.956	0.958	0.955	0.959
	(0.011)	(0.013)	(0.016)	(0.011)	(0.020)	(0.007)
N	14,808	5,847	8,269	6,809	10,664	5,815
Panel C. Per capita income						
Average pass-through	0.868	0.918	0.962	0.948	0.940	0.936
	(0.051)	(0.033)	(0.021)	(0.014)	(0.016)	(0.018)
N	11,987	5,222	6,562	7,106	8,007	4,490
Panel D. Two-way-fixed effects						
Average pass-through	0.869	0.879	0.937	1.004	0.998	0.974
	(0.061)	(0.033)	(0.027)	(0.036)	(0.035)	(0.038)
N	121,728	34,974	51,249	46,905	77,686	93,619

Notes: Estimates are average pass-through rates of tax changes into retail prices (ρ; cent/liter price change per cent/liter tax change) obtained via Equations 5-7, for each of six income groups. Each panel provides results with a different deviation from the base specification (whose results are presented in Table 3, Panel A), which uses station-year-month observations weighted by station-year diesel sales, median-income quintiles, a twelve-month post-period, and province-level clustering of standard errors. Q1-Q5 indicate quintiles of the municipality median (or per-capita) income distribution; "No Inc" denotes the group of stations in towns too small to be included in the data. 'N' reports the number of treatment-group post-period observations used in Panels A-C and the number of total observations used in Panel D.

plots decile-specific pass-through estimates and confidence intervals. It reveals a steady upward pass-through trend in municipality income decile. I include, at income decile "0" in the figure, the pass-through rate measured in the group of rural stations with no income data. To the extent that such towns have the lowest incomes, the "Q0" rate contributes further to the clear pattern of pass-through progressivity observable from Q1 to Q10.

The rates plotted in Figure 9 are based on municipality median income rather than household income. I stop short of incorporating these decile-specific rates directly into Figure 8 because I do

Figure 8: *Household auto fuel consumption relative to total expenditure*



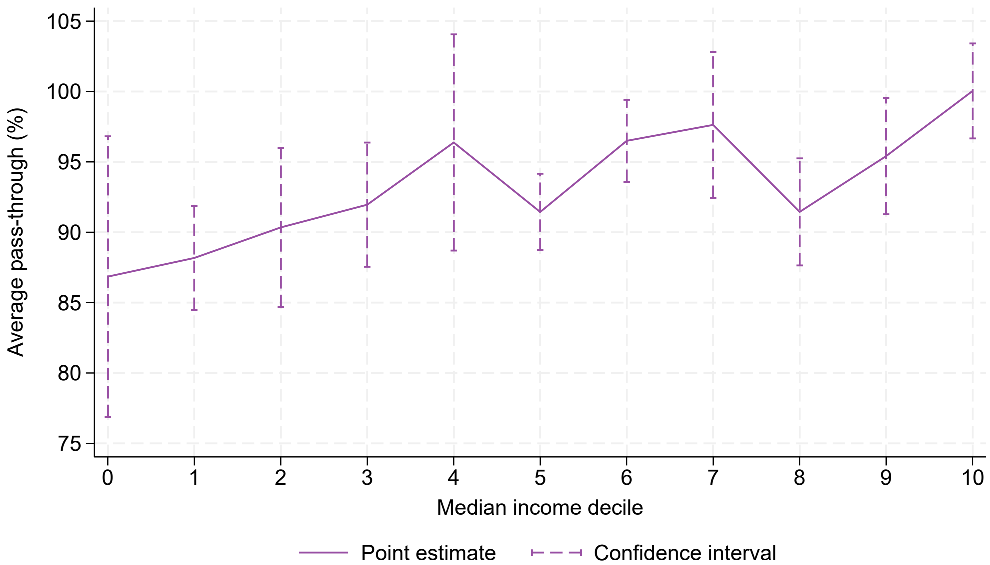
Notes: Data are from the Spanish Household Budget Survey, omitting regions and territories that are not in the main analysis sample. I apply survey weights for national representativeness.

not observe the municipality of households in the budget survey. Thus I cannot match pass-through rates to households without the strong assumption that quantile of municipality income and quantile of household income are equal.[20] There are clearly differences *within* municipality in fuel prices, propensity to commute, and willingness to search. Thus, municipal-average pass-through rates may not be an accurate predictor of household-specific pass-through. However, consumers' willingness to search is theoretically falling in income due to a higher monetary cost of time among the wealthy. All else equal, then, relatively poorer households are more likely to search for and find a lower price than the municipality average. This logic supports the notion that the progressive pass-through finding holds even at the higher resolution of household-level income data.

Comparison of slopes in Figures 8 and 9 can shed light on the potential size of the progressive pass-through effect on distributional welfare. The proportional drop in consumption with rising expenditure decile (in Figure 8) is greater than the proportional increase in pass-through (Figure 9). This implies that the consumer surplus impacts of these tax changes are overall regressive in spite of progressive pass-through. If the pass-through patterns I identify were to hold across the national household wealth distribution, the effect would be a dampening of the apparent regressivity in Figure 8. The magnitude of this potential dampening is likely modest: households in the bottom

[20]The potential for variable demand response also remains a fundamental problem for the simplified distributional welfare calculation. A higher demand elasticity means lower pass-through, which reduces the consumer surplus loss of the tax – but it also means more foregone consumption, which *raises* the consumer surplus loss. Given that retail auto fuel demand tends to be relatively inelastic (though less than previously thought; see Kilian and Zhou, 2024), the former effect likely dominates the latter, but incorporating variable pass-through without variable demand response is still potentially misleading. Harju et al. (2022) work around this issue by assuming that demand is perfecly inelastic, but they acknowledge that their distributional welfare exercise should only be considered an illustration, rather than definitive findings.

Figure 9: *Average pass-through by income decile*



Notes: Data points are twelve-month average pass-through rates of tax changes (ρ; cent/liter price change per cent/liter tax change) and associated confidence intervals, obtained via Equations 5-7 separately for each decile of the municipality median income distribution as well as the group of stations without income data (corresponding to $x = 0$). Observations are station-year-months and are weighted by station-year diesel quantity sold. Standard errors are clustered by province.

expenditure decile buy 25-60 percent more fuel per dollar total expenditure than those in the top decile (Figure 8), while the corresponding bottom-decile pass-through rate (Figure 9) is only about 13 percent smaller.

5.2 Causal mechanisms

The descriptive fact that fuel tax pass-through is greater in higher-income municipalities is what matters for distributional welfare in the Spanish context. However, the *cause* of this pattern is important as well, especially because it can inform external validity considerations. The causes of my empirical finding suggest which attributes of other countries matter in assessing the potential for external validity. If higher pass-through in wealthier areas is being driven by wealthier areas having more inelastic demand, then the demand/wealth relationship in other countries is what matters most. On the other hand, if other aspects of auto fuel markets – such as market structure, supply chains, or geography – drive the results, then it is countries with similarities in these particular attributes that one might argue for external validity.

The dCDH method does not accommodate interaction terms with pass-through, so I cannot estimate relationships between pass-through and several characteristics simultaneously like is commonly done with TWFE regression. Instead I replicate the estimation of pass-through by income group in subgroups defined by salient attributes. I focus on station branding, since pass-through varies most significantly with this attribute. Table 5 displays pass-through estimates by income group in each of two different brand groups: refiner (Panel A) and independent (Panel B). Column 1 reveals that average pass-through is significantly higher at refiner-branded stations (95.3 percent) than at independent-branded ones (82.2 percent). Columns 2 through 7 show that this cross-brand gradient holds within every income group. The gap between refiner brands and independent brands is particularly large in the no-income group (column 2) – 94.4 versus 64.1 percent – and the low rate among independent brands explains the low rate overall in this income group. Meanwhile, both brand types are associated with low pass-through in income quintile 1 (column 3).

Table 5: *Pass-through and station branding*

	All (1)	No Inc (2)	Q1 (3)	Q2 (4)	Q3 (5)	Q4 (6)	Q5 (7)
Panel A. Refinery-branded stations							
Average pass-through	0.953	0.944	0.848	0.967	0.961	0.959	0.974
	(0.008)	(0.015)	(0.018)	(0.021)	(0.012)	(0.016)	(0.008)
N	29,492	7,767	3,310	4,713	4,097	6,274	3,319
Panel B. Independent-branded stations							
Average pass-through	0.822	0.641	0.829	0.898	0.862	0.944	0.867
	(0.055)	(0.143)	(0.028)	(0.040)	(0.034)	(0.081)	(0.078)
N	8,674	3,056	535	1,544	924	1,270	608

Notes: Point estimates are average pass-through rates of tax changes into retail prices (ρ; cent/liter price change per cent/liter tax change), estimated via Equations 5-7. In Panel A, only stations bearing the brand of an oil refiner in Spain are considered; in Panel B, only stations with an *independent* brand are considered. Column 1 displays overall average pass-through rates by brand type, and columns 2-6 display brand-specific rates by income grouping (quintiles $Q1$-$Q5$ of the municipality median income distribution and the group of stations without income data). All specifications use station-year-month observations, weighted by station-year diesel quantity sold, with a twelve-month post-period and standard errors clustered by province. 'N' is the number of treatment-group post-period observations used.

I inspect several other station attributes similarly and present the results in the Appendix. Point estimate magnitudes suggest that pass-through is lower among: (a) stations with no local brand concentration (Appendix Table A7); (b) stations in the top quintile of average sales volume (Appendix Table A8); and (c) stations in the bottom quintile of municipal population density (Appendix Figure A9).[21] The pass-through gradient in income among subgroups (a) and (b), in addition to the independent-station subgroup, remains similar to that of the full station sample. Meanwhile, number of nearby rivals – another measure of local competition – has no consistent relationship wih pass-through (Appendix Table A9, Panel A), and among refiner-branded stations, neither does vertical

[21] There is not enough income-quintile variation *within* population-density quintile to construct the analogous version of Table 5 – this is because municipalities in the bottom quintile of population density are almost entirely municipalities that are missing income by virtue of having population less than 5,000.

integration (Appendix Table A9, Panel B). These results collectively suggest that market power, vertical structure, and the presence of public transit alternatives (to the extent that high population density predicts this) do not fully explain progressive pass-through up the municipality income distribution. Across all groups inspected, independent stations in the most rural areas have the lowest average pass-through rate (64.1 percent), but the cause of this result is unclear.

5.3 Pass-through of other fuel costs

Taxes are nearly half of retail prices in this time period in Spain; the production cost of fuel is the other major source of variable costs of auto fuel supply. The fact that fuel tax pass-through rises in income thus begs the question of whether *non*-tax fuel cost pass-through does as well. Tax pass-through may differ from non-tax cost pass-through because of an underlying difference in the fuel elasticity of demand with respect to each of these cost types. There is ongoing academic interest in whether such a difference in elasticities exists, because this dictates whether one can use either of the estimates to proxy for the other – for instance, in a forecast of the impacts of a hypothetical tax change. Many researchers have found that the two elasticites do differ (e.g., Davis and Kilian, 2011; Li et al., 2014), but Kilian and Zhou (2024) is a recent, notable exception in the US context. Li et al. (2014) propose that such a difference in elasticities may occur because gas taxes are more persistent than (e.g.) the price of crude oil and also possibly more salient to the consumer.

I test for heterogeneity in crude oil price pass-through by income group using the same regression specification as Kilian and Zhou (2024):

$$\Delta P_{it} = \alpha + \beta \Delta \text{Crude}_t + \theta_y + \nu_m + \phi_i + \epsilon_{it}, \tag{8}$$

with fixed effects by year θ_y and by month ν_m separately rather than year-month t, to accommodate the time-series of crude oil price as the "treatment" variable. ΔP and ΔCrude are changes in pre-tax retail price and Brent crude oil price from the previous month to the current one. Table 6 documents the income group specific pass-through estimates. It is apparent that crude price pass-through, at 99-100 percent, is higher than retail tax pass-through (85-95 percent). While very small standard errors make some of the differences across income group statistically significant, the magnitudes of these differences are in the range of 0-1 percentage points. Other specifications of the crude price pass-through regression yield qualitatively similar results (these are available upon request). This suggests that there *is* a difference here between the consumer response to an auto fuel tax versus an oil price shock.

6 Conclusion

Automotive fuel taxes are used in dozens of countries around the world, and energy taxes more generally are a key current and future policy lever for both driving reductions in fossil fuel use and raising public revenues. The distributional equity of such policies is a fundamental political and moral issue that cannot be assessed by theory alone. In this paper, I contribute to our understanding of the distributional impacts of energy taxation by studying the empirical relationship between

Table 6: *Average crude price pass-through by income group*

	No Inc	Q1	Q2	Q3	Q4	Q5
	(1)	(2)	(3)	(4)	(5)	(6)
1-month Δ Brent crude price	0.996	0.986	0.991	0.997	0.994	0.998
	(0.004)	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)
N	119,689	34,397	50,435	46,158	76,449	92,118

Notes: Point estimates are average pass-through rates of one-month changes in the Brent benchmark crude oil price to one-month retail price changes (cent/liter retail price change per cent/liter crude price change), estimated via Equation 8, separately for each quintile of the municipality median income distribution ($Q1$-$Q5$) as well as the group of stations without income data ("No Inc""). Observations are station-year-months and are weighted by station-year diesel quantity sold. Standard errors are clustered by province. 'N' is the number of treatment-group post-period observations used.

auto fuel tax pass-through and income. If pass-through varies with wealth, it has consequences for distributional welfare that are not generally accounted for in welfare analysis.

I find that a regional auto fuel tax in Spain, which was increased 14 times in staggered fashion across 11 regions from 2010-2012, exhibits variable pass-through in income. While a central estimate of national average pass-through 92 percent, estimates of pass-through at the bottom of the municipal median income distribution are significantly lower. In particular, the bottom-quintile group and the group of smallest towns lacking income data are associated with with 7-11 percentage-point reductions in pass-through relative to the other quintiles. Decile-specific estimates show pass-through following a steady upward trend in income level. This finding of "progressive" pass-through stands in contrast to the one other existing study of energy tax pass-through and income (Harju et al., 2022), but both results are possible because of the many causes of heterogeneous pass-through and the varying empirical contexts being studied.

My findings illustrate the relevance of heterogeneous pass-through to distributional welfare analysis. A typical distributional welfare calculation for the Spanish auto fuel tax, in assuming uniform pass-through, would yield results that are likely biased towards greater regressivity. This study also provides an early example of the application of one of the newest difference-in-differences methods, from de Chaisemartin and D'Haultfoeuille (2024a), to facilitate unbiased treatment effect estimation in settings with staggered, non-binary treatments. Future research efforts may profitably focus on improving the wealth measure by which to separately estimate pass-through rates (income is an imperfect proxy for lifetime wealth, and municipal measures are not the same as household ones), and in determining the causes of documented pass-through/wealth correlations. Amid an ongoing global transition to electric vehicles (EVs), research on price elasticities of demand as well as energy cost pass-through for EV drivers will be especially policy relevant.

References

ALM, J., E. SENNOGA, AND M. SKIDMORE (2009): "Perfect Competition, Urbanization, and Tax Incidence in the Retail Gasoline Market." *Economic Inquiry*, 47, 118–134.

ATKIN, D. AND D. DONALDSON (2015): "Who's Getting Globalized? The Size and Implications of Intra-national Trade Costs." Working Paper.

BAJO-BUENESTADO, R. AND M. BORRELLA-MAS (2022): "The Heterogeneous Tax Pass-through under Different Vertical Relationships," *Economic Journal*, 132, 1684–1708.

BENTO, A., L. GOULDER, M. JACOBSEN, AND R. VON HAEFEN (2009): "Distributional and Efficiency Impacts of Increased US Gasoline Taxes." *American Economic Review*, 99, 667–699.

BONNET, C., P. DUBOIS, S. VILLAS BOAS, AND D. KLAPPER (2013): "Empirical Evidence on the Role of Nonlinear Wholesale Pricing and Vertical Restraints on Cost Pass-through." *Review of Economics and Statistics*, 95, 500–515.

BORENSTEIN, S., A. CAMERON, AND R. GILBERT (1997): "Do Gasoline Prices Respond Asymmetrically to Crude Oil Price Changes?" *Quarterly Journal of Economics*, X, 306–339.

BORUSYAK, K., X. JARAVEL, AND J. SPIESS (2024): "Revisiting Event-Study Designs: Robust and Efficient Estimation." *Review of Economic Studies*, 1–33.

CALLAWAY, B. AND P. SANT'ANNA (2021): "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics*, 225, 200–230.

CAMERON, A. AND D. GILBERT (2015): "A Practitioner's Guide to Cluster-Robust Inference." *Journal of Human Resources*, 50, 317–372.

CASPERSEN, E. AND G. METCALF (1994): "Is a Value Added Tax Regressive? Annual Versus Lifetime Incidence Measures." *National Tax Journal*, 47, 731–746.

CAVALLO, A., G. GOPINATH, B. NEIMAN, AND J. TANG (2021): "Tariff Pass-through at the Border and at the Store: Evidence from US Trade Policy." *American Economic Review: Insights*, 3, 19–34.

CHETTY, R. (2009): "Sufficient Statistics for Welfare Analysis: A Bridge Between Structural and Reduced-Form Methods." *Annual Review of Economics*, 1, 451–487.

CONTÍN-PILART, I., A. CORRELJÉ, AND M. PALACIOS (2009): "Competition, Regulation, and Pricing Behaviour in the Spanish Retail Gasoline Market." *Energy Policy*, 37, 219–228.

COSTA-FONT, J. AND A. RICO (2006): "Devolution and the Interregional Inequalities in Health and Healthcare in Spain," *Regional Studies*, 875–887.

COURT OF JUSTICE OF THE EUROPEAN UNION (2014): "Press Release: The Tax on Retail Sales of Certain Hydrocarbons is Contrary to European Union Law," .

DAVIS, L. AND L. KILIAN (2011): "Estimating the Effect of a Gasoline Tax on Carbon Emissions." *Journal of Applied Econometrics*, 62, 1187–1214.

de Chaisemartin, C., D. Ciccia, X. D'Haultfoeuille, F. Knau, M. Malézieux, and D. Sow (2024): "Event Study Estimators of Variance Estimators Computed by the did_multiplegt_dyn Command." .

de Chaisemartin, C. and X. D'Haultfoeuille (2020): "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review*, 110, 2964–2996.

——— (2023): "Two-Way Fixed Effects and Differences-in-Differences with Heterogeneous Treatment Effects: A Survey." *Econometrics Journal*, 26, C1–C30.

——— (2024a): "Difference-in-Differences Estimators of Intertemporal Treatment Effects." *Review of Economics and Statistics*, 1–45.

——— (2024b): "Stata help file: did_multiplegt," .

Doyle Jr., J. and K. Samphantharak (2008): "$2.00 Gas! Studying the Effects of a Gas Tax Moratorium." *Journal of Public Economics*, 92, 869–884.

Fabra, N. and M. Reguant (2014): "Pass-through of Emissions Costs in Electricity Markets." *American Economic Review*, 104, 2872–2899.

Flues, F. and Thomas, A. (2015): "The Distributional Effects of Energy Taxes." Tech. rep., Organization for Economic Cooperation and Development, <https://www.oecd-ilibrary.org/the-distributional-effects-of-energy-taxes_5js1qwkqqrbv.pdf?itemId=%2Fcontent%2Fpaper%2F5js1qwkqqrbv-en&mimeType=pdf>.

Ganapati, S., J. Shapiro, and R. Walker (2020): "Energy Cost Pass-Through in US Manufacturing: Estimates and Implications for Carbon Taxes." *American Economic Journal: Applied Economics*, 12, 303–342.

Genakos, C. and M. Pagliero (2022): "Competition and Pass-through: Evidence from Isolated Markets." *American Economic Journal: Applied Economics*, 14, 35–57.

Goodman-Bacon, A. (2021): "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics*, 225, 254–277.

Gruber, J. and B. Koszegi (2004): "Tax Incidence When Individuals are Time-Inconsistent: The Case of Cigarette Excise Taxes." *Journal of Public Economics*, 88, 1959–1987.

Harju, J., T. Kosonen, M. Laukkanen, and K. Palanne (2022): "The Heterogeneous Incidence of Fuel Carbon Taxes: Evidence from Station-Level Data." *Journal of Environmental Economics and Management*, 112, 102607.

Horowitz, J., J.-A. Cronin, H. Hawkins, L. Konda, and A. Yuskavage (2017): "Methodology for Analyzing Carbon Taxes." Tech. rep., United States Department of Treasury, Office of Tax Analysis, <https://home.treasury.gov/system/files/131/WP-115.pdf>.

Houde, J.-F. (2012): "Spatial Differentiation and Vertical Mergers in Retail Markets for Gasoline." *American Economic Review*, 102, 2147–2182.

Hughes, J., C. Knittel, and D. Sperling (2008): "Evidence of a Shift in the Short-Run Price Elasticity of Gasoline Demand." *Energy Journal*, 29, 113–134.

Jaffe, S. and E. Weyl (2013): "The First-Order Approach to Merger Analysis." *American Economic Journal: Microeconomics*, 5, 188–218.

Jenkin, H. (1872): "On the Principles that Regulate the Incidence of Taxes." *Proceedings of the Royal Society of Edinburgh*, 7, 618–631.

Kayser, H. (2000): "Gasoline Demand and Car Choice: Estimating Gasoline Demand Using Household Information," *Energy Economics*, 22, 331–348.

Kilian, L. and X. Zhou (2024): "Heterogeneity in the Pass-through from Oil to Gasoline Prices: A New Instrument for Estimating the Price Elasticity of Gasoline Demand." *Journal of Public Economics*, 232, 105099.

Kroft, K., J.-W. Laliberte, R. Leal-Vizcaino, and M. Notowidigdo (2024): "Salience and Taxation with Imperfect Competition," *Review of Economic Studies*, 91, 403–437.

Lade, G. and J. Bushnell (2022): "Fuel Subsidy Pass-Through and Market Structure: Evidence from the Renewable Fuel Standard." *Journal of the Association of Environmental and Resource Economists*, 6, 563–592.

Li, S., J. Linn, and E. Muehlegger (2014): "Gasoline Taxes and Consumer Behavior." *American Economic Journal: Economic Policy*, 6, 302–342.

Marion, J. and E. Muehlegger (2011): "Fuel Tax Incidence and Supply Conditions." *Journal of Public Economics*, 95, 1202–1212.

Marshall, A. (1890): *Principles of Economics.*, New York: Macmillan.

Mathur, A. and A. Morris (2012): "Distributional Effects of a Carbon Tax in Broader U.S. Fiscal Reform." Tech. rep., Brookings Institute.

Metcalf, G. (1999): "A Distributional Analysis of Green Tax Reforms," *National Tax Journal*, 52, 655–681.

Miller, N., M. Osborne, and G. Sheu (2017): "Pass-through in a Concentrated Industry: Empirical Evidence and Regulatory Implications." *RAND Journal of Economics*, 48, 69–93.

Muehlegger, E. and D. Rapson (2022): "Subsidizing Low- and Middle-Income Adoption of Electric Vehicles: Quasi-experimental Evidence from California," *Journal of Public Economics*, 216, 104752.

Muehlegger, E. and R. Sweeney (2022): "Pass-through of Own and Rival Cost Shocks: Evidence from the US Fracking Boom," *Review of Economics and Statistics*, 104, 1361–1369.

National Commission on Competition (2012): "Report on the Consultation Request Submitted by the Secretary of State for the Economy and Support for Enterprises Regarding the Automotive Fuel Market in Spain." Tech. rep., CNC, España, <https://www.cnmc.es/sites/default/files/1186976_7.pdf>.

National Commission on Markets and Competition (2024): "Boletín Informativo de la Distribución de Carburantes en Estaciones de Servicio." Tech. rep., CNC, España, <https://www.cnmc.es/sites/default/files/5218929.pdf>.

Noceda, M. and M. Jimenez (2015): "Competition Authorities Fine Five Oil Companies 32.4 Million Euros for Price Fixing." *El País*, <https://elpais.com/economia/2015/02/25/actualidad/1424859682_727919.html>.

Perdiguero, J. and J. Borrell (2007): "La Difícil Conducción de la Competencia por el Sector de las Gasolineras en España." *Economía Industrial*, 365, 113–125.

Pizer, W. and S. Sexton (2020): "The Distributional Impacts of Energy Taxes." *Review of Environmental Economics and Policy*, 13, 104–123.

Pless, J. and A. van Benthem (2019): "Pass-through as a Test for Market Power: An Application to Solar Subsidies." *American Economic Journal: Applied Economics*, 11, 367–401.

Poterba, J. (1991): "Is the Gasoline Tax Regressive?" *Tax Policy and the Economy*, 5, 145–164.

——— (1996): "Retail Price Reactions to Changes in State and Local Sales Taxes." *National Tax Journal*, 49, 165–176.

Rico-Straffon, J., Z. Wang, S. Panlasigui, C. Loucks, J. Swenson, and A. Pfaff (2023): "Forest Concessions and Eco-certifications in the Peruvian Amazon: Deforestation Impacts of Logging Rights and Logging Restrictions." *Journal of Environmental Economics and Management*, 118, 102780.

Roth, J. (2022): "Pretest with Caution: Event-Study Estimates after Testing for Parallel Trends." *American Economic Review: Insights*, 4, 305–322.

Spanish Tax Agency (2024): "Devolución especial IVMDH: Preguntas Frecuentes," .

Sterner, T. (2012): "Fuel Taxes, Climate and Tax Incidence." *Fuel Taxes and the Poor: The Distributional Effects of Gasoline Taxation and Their Implications for Climate Policy*.

Stolper, S. (2016): "Competition and Incidence: Automotive Fuel Tax Pass-through at State Borders," Working Paper.

Sun, L. and S. Abraham (2021): "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics*, 225, 175–199.

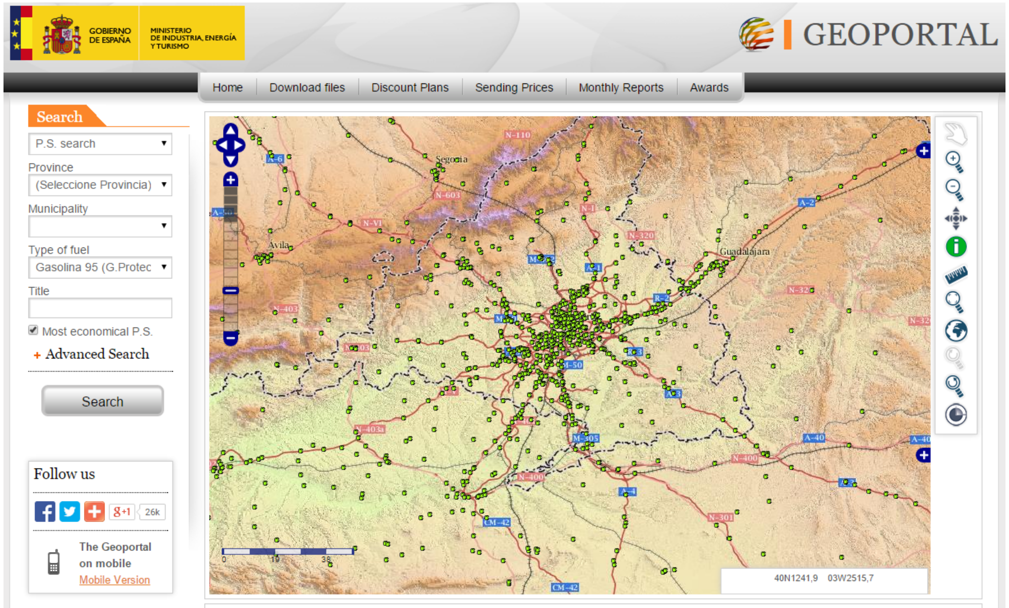
West, S. (2004): "Distributional Effects of Alternative Vehicle Pollution Control Policies." *Journal of Public Economics*, 88, 735–757.

Weyl, E. and M. Fabinger (2013): "Pass-through as an Economic Tool: Principles of Incidence under Imperfect Competition." *Journal of Political Economy*, 121, 528–583.

Wing, C., M. Yozwiak, A. Hollingsworth, S. Freedman, and K. Simon (2024): "Designing Difference-in-DIfference Studies with Staggered Treatment Adoption: Key Concepts and Practical Guidelines." *Annual Review of Public Health*, 45, 485–505.

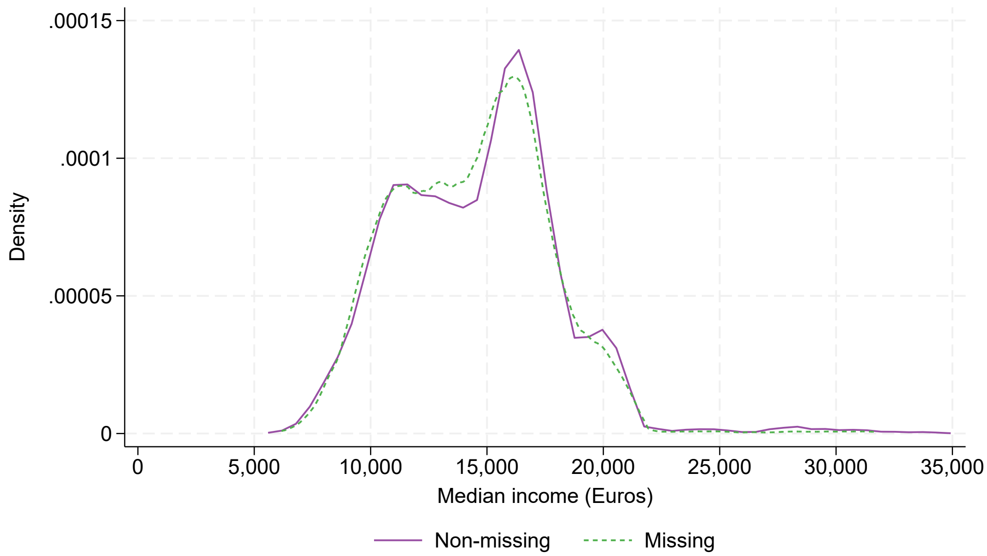
Appendix to "Income and Energy Tax Pass-through: Evidence from Gas Stations" (Stolper, 2025).

Figure A1: *Geoportal screenshot*



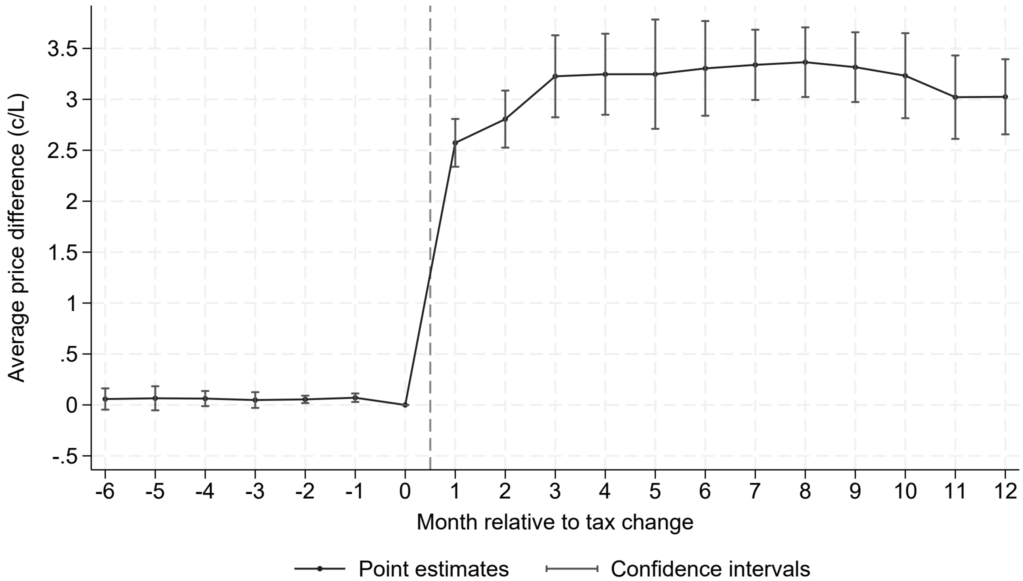
Notes: Green dots are Spanish retail automotive fuel stations. The screenshot shows the Madrid metro area. Source: <https://geoportalgasolineras.es/geoportal-instalaciones/Inicio>, accessed on February 15th, 2015.

Figure A2: *Income distribution of stations, by sales data availability*



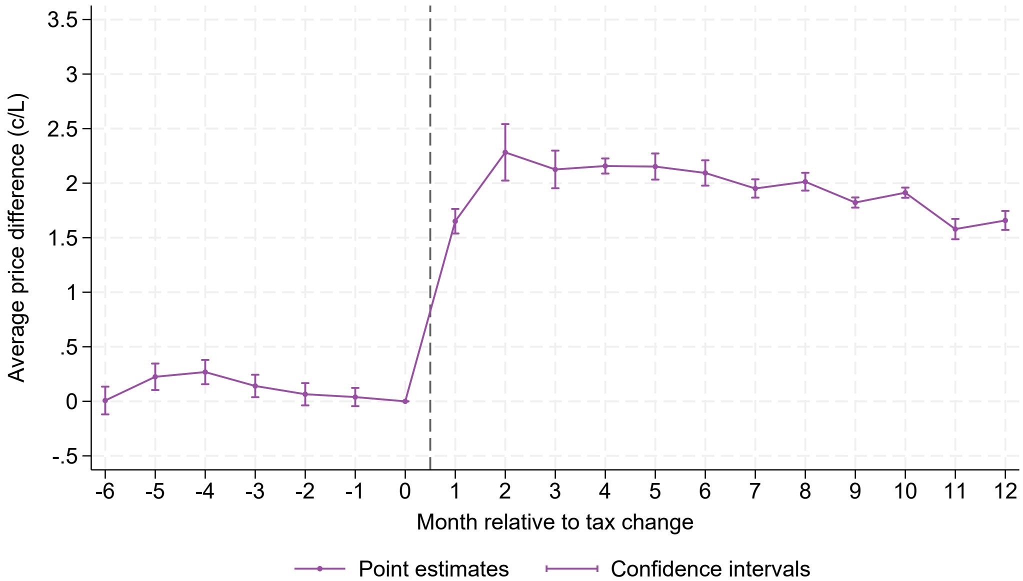
Notes: Plotted curves are Epanechnikov kernel densities of average municipality median income from 2008-2013 among stations missing versus not missing diesel sales data. The y-axis measures estimated probability density at the given value on the x-axis.

Figure A3: *Average price impacts over time, no-income group*



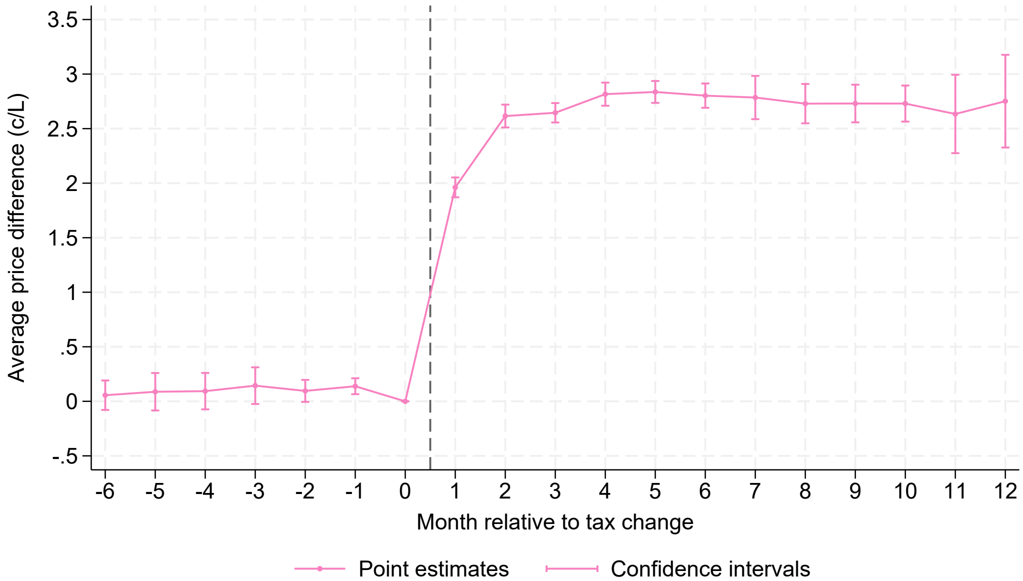
Notes: The sample is the group of stations in small towns (population less than 5,000) lacking income data. Point estimates and confidence intervals are for monthly DD coefficients (DD_ℓ; cent/liter price change per cent/liter tax change) estimated using Equations 5 and 6 and a monthly analog of Equation 7, where $\ell \in [-6, 12]$ and month zero is omitted. Observations are station-year-months and are weighted by station-year diesel quantity sold. Standard errors are clustered by province.

Figure A4: *Average price impacts over time, 1st income quintile*



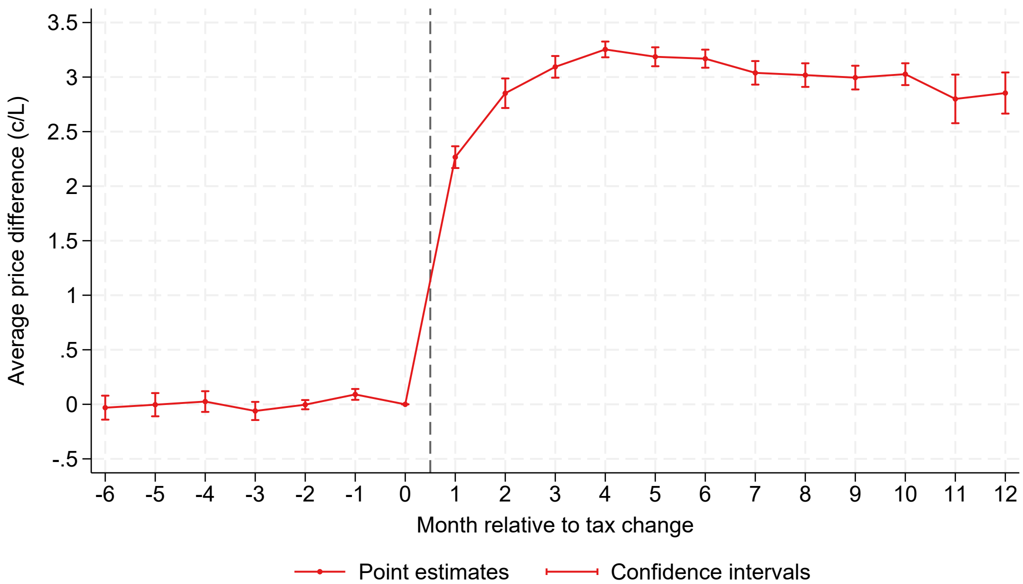
Notes: The sample is the group of stations in the bottom quintile of the municipal median income distribution. Point estimates and confidence intervals are for monthly DD coefficients (DD_ℓ; cent/liter price change per cent/liter tax change) estimated using Equations 5 and 6 and a monthly analog of Equation 7, where $\ell \in [-6, 12]$ and month zero is omitted. Observations are station-year-months and are weighted by station-year diesel quantity sold. Standard errors are clustered by province.

Figure A5: *Average price impacts over time, 2nd income quintile*



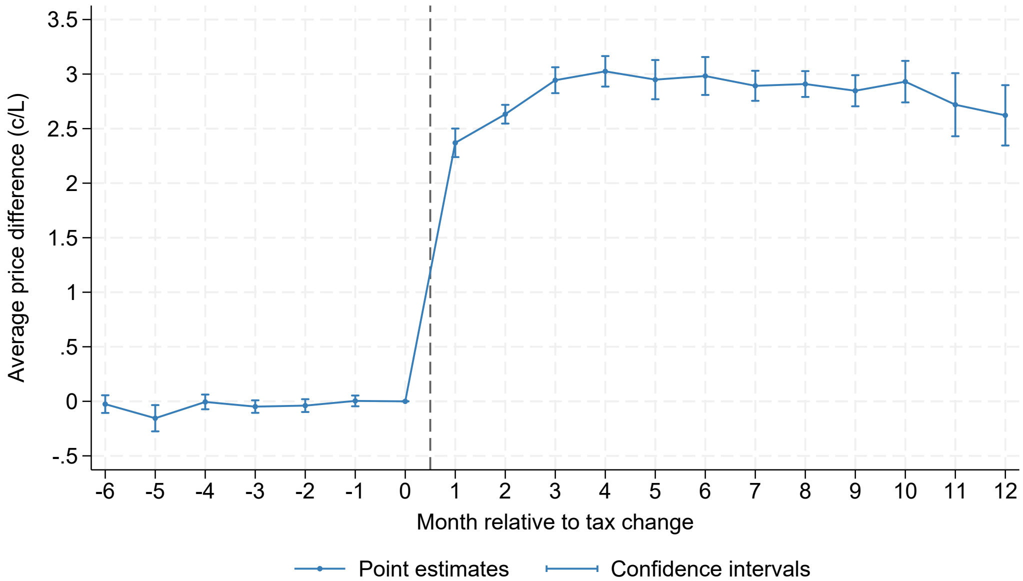
Notes: The sample is the group of stations in the second quintile of the municipal median income distribution. Point estimates and confidence intervals are for monthly DD coefficients (DD_ℓ; cent/liter price change per cent/liter tax change) estimated using Equations 5 and 6 and a monthly analog of Equation 7, where $\ell \in [-6, 12]$ and month zero is omitted. Observations are station-year-months and are weighted by station-year diesel quantity sold. Standard errors are clustered by province.

Figure A6: *Average price impacts over time, 3rd income quintile*



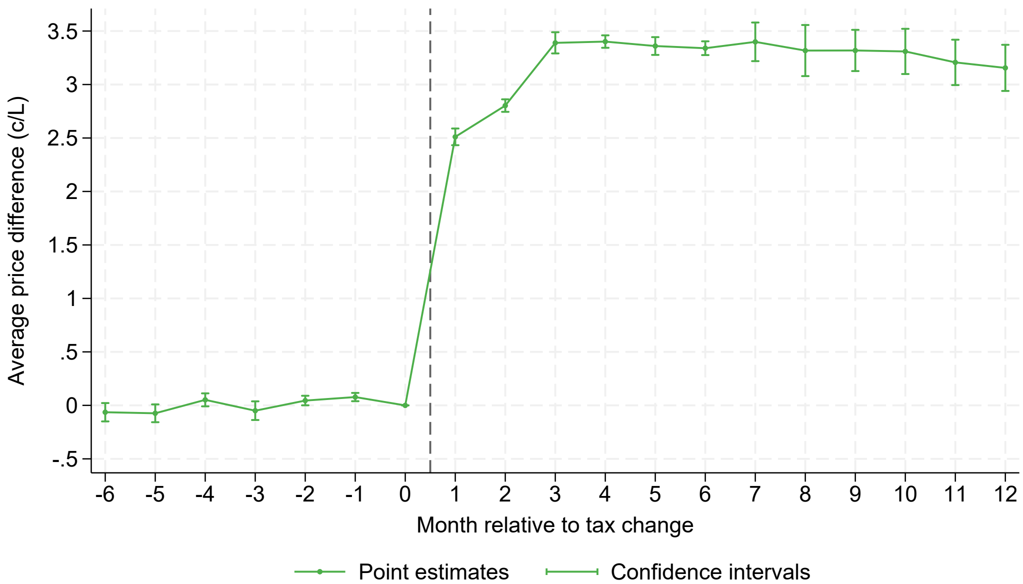
Notes: The sample is the group of stations in the third quintile of the municipal median income distribution. Point estimates and confidence intervals are for monthly DD coefficients (DD_ℓ; cent/liter price change per cent/liter tax change) estimated using Equations 5 and 6 and a monthly analog of Equation 7, where $\ell \in [-6, 12]$ and month zero is omitted. Observations are station-year-months and are weighted by station-year diesel quantity sold. Standard errors are clustered by province.

Figure A7: *Average price impacts over time, 4th income quintile*



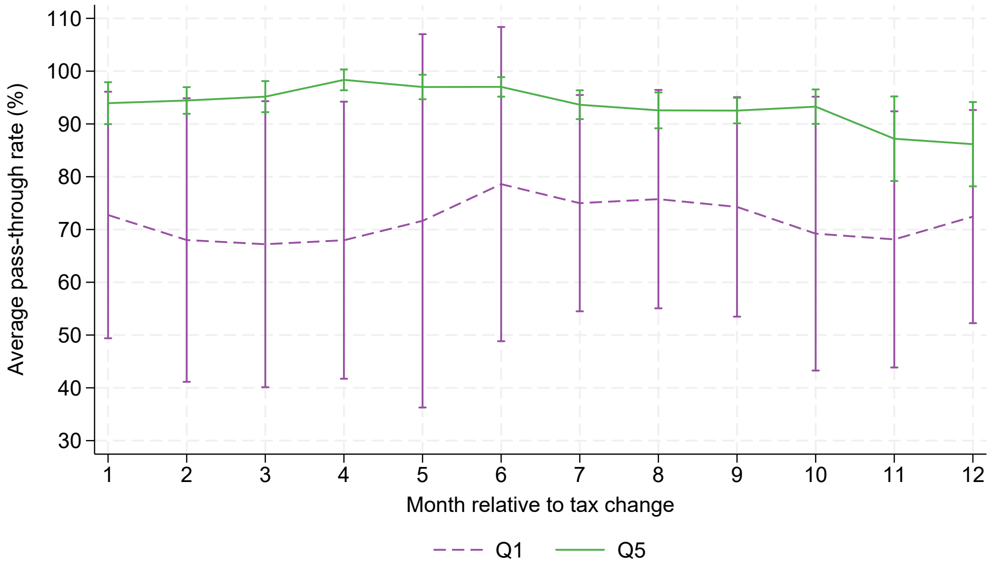
Notes: The sample is the group of stations in the fourth quintile of the municipal median income distribution. Point estimates and confidence intervals are for monthly DD coefficients (DD_ℓ; cent/liter price change per cent/liter tax change) estimated using Equations 5 and 6, where $\ell \in [-6, 12]$ and month zero is omitted. Observations are station-year-months and are weighted by station-year diesel quantity sold. Standard errors are clustered by province.

Figure A8: *Average price impacts over time, 5th income quintile*



Notes: The sample is the group of stations in the fifth quintile of the municipal median income distribution. Point estimates and confidence intervals are for monthly DD coefficients (DD_ℓ; cent/liter price change per cent/liter tax change) estimated using Equations 5 and 6 and a monthly analog of Equation 7, where $\ell \in [-6, 12]$ and month zero is omitted. Observations are station-year-months and are weighted by station-year diesel quantity sold. Standard errors are clustered by province.

Figure A9: *Pass-through by population density*



Notes: Lines are constructed from monthly DD point estimates of average pass-through rate (ρ; cent/liter price change per cent/liter tax change) by municipal population density quintile, estimated using Equations 5 and 6 and a monthly analog of Equation 7, where $\ell \in [1, 12]$. Vertical lines are confidence intervals. 'Q1' and 'Q5' denote quintiles 1 and 5, respectively. Observations are station-year-months and are weighted by station-year diesel quantity sold. Standard errors are clustered by province.

Table A1: *Summary statistics, weighted by station sales*

	All	No Inc	Q1	Q2	Q3	Q4	Q5
Retail diesel price (c/L)	100.39	99.93	100.07	100.31	100.38	100.49	100.84
	(2.94)	(3.16)	(2.83)	(2.72)	(2.87)	(3.03)	(2.76)
Regional retail tax (c/L)	1.92	1.83	1.76	1.85	1.81	1.91	2.10
	(0.86)	(1.04)	(0.57)	(0.66)	(0.82)	(0.79)	(0.85)
Refiner-branded (0/1)	0.68	0.65	0.71	0.69	0.67	0.68	0.70
	(0.47)	(0.48)	(0.45)	(0.46)	(0.47)	(0.47)	(0.46)
Wholesaler-branded (0/1)	0.12	0.11	0.11	0.11	0.11	0.12	0.14
	(0.32)	(0.31)	(0.31)	(0.31)	(0.31)	(0.33)	(0.35)
Independent-branded (0/1)	0.20	0.25	0.18	0.20	0.22	0.20	0.16
	(0.40)	(0.43)	(0.38)	(0.40)	(0.41)	(0.40)	(0.36)
Brand vertically integrated (0/1)	0.29	0.25	0.16	0.22	0.25	0.31	0.37
	(0.45)	(0.44)	(0.37)	(0.41)	(0.43)	(0.46)	(0.48)
# rivals, distance-weighted	1.52	0.82	0.72	1.05	1.40	1.89	2.30
	(1.38)	(1.01)	(0.74)	(1.02)	(1.24)	(1.32)	(1.51)
Proportion of rivals sharing one's brand	0.28	0.37	0.26	0.25	0.25	0.24	0.25
	(0.30)	(0.40)	(0.34)	(0.29)	(0.29)	(0.24)	(0.23)
Municipality population (1,000s)	204.45	2.05	17.48	32.52	32.91	141.95	620.63
	(613.14)	(1.35)	(17.60)	(45.03)	(30.57)	(155.06)	(1,066.90)
Municipality pop. density (1,000s per km2)	1.32	0.06	0.16	0.32	0.90	1.60	3.07
	(2.61)	(0.14)	(0.22)	(0.68)	(1.74)	(2.87)	(3.50)
Mean of median income across years	15,164.04	.	9,421.58	11,365.05	13,232.76	15,459.85	18,731.81
	(3,615.73)	(.)	(844.02)	(529.47)	(514.89)	(628.04)	(2,875.05)
Observations	7,564	2,245	634	896	816	1,354	1,619

Notes: Sample means, weighted by total station-level diesel quantity sold in the observation period, and standard deviations (in parentheses) are calculated from station-level observations. "All" refers to all stations in the main analysis sample. "No Inc" indicates stations in municipalities too small to have income data reported. *Q*1 through *Q*5 denote quintiles of the municipality median income distribution. Brand vertically integrated stations are owned and operated by a wholesaler or refiner. Proportion of rivals sharing one's brand is calculated within a ten-minute radius of each station and takes a value of one in the case of zero rivals. # of rivals is similarly calculated within a ten-minute radius and is a weighted sum, with weights equal to inverse travel time in minutes to each rival.

Table A2: *Summary statistics on additional station attributes*

	All	No Inc	Q1	Q2	Q3	Q4	Q5
Vertically integrated (0/1)	0.24	0.15	0.13	0.21	0.28	0.31	0.35
	(0.43)	(0.36)	(0.34)	(0.41)	(0.45)	(0.46)	(0.48)
Vertically independent (0/1)	0.21	0.28	0.25	0.22	0.19	0.17	0.15
	(0.41)	(0.45)	(0.44)	(0.41)	(0.40)	(0.37)	(0.35)
Commission contract (0/1)	0.33	0.38	0.39	0.34	0.30	0.30	0.27
	(0.47)	(0.49)	(0.49)	(0.47)	(0.46)	(0.46)	(0.44)
Firm contract (0/1)	0.18	0.16	0.19	0.19	0.20	0.19	0.20
	(0.39)	(0.37)	(0.39)	(0.39)	(0.40)	(0.39)	(0.40)
Lavado	0.46	0.37	0.57	0.51	0.51	0.49	0.47
	(0.50)	(0.48)	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)
AguaAire	0.66	0.67	0.75	0.70	0.63	0.61	0.62
	(0.48)	(0.47)	(0.43)	(0.46)	(0.48)	(0.49)	(0.48)
Cafeteria	0.17	0.20	0.19	0.15	0.16	0.16	0.14
	(0.38)	(0.40)	(0.39)	(0.36)	(0.37)	(0.37)	(0.35)
Tienda	0.68	0.66	0.72	0.73	0.68	0.68	0.68
	(0.47)	(0.47)	(0.45)	(0.45)	(0.47)	(0.47)	(0.47)
Observations	7,564	2,245	634	896	816	1,354	1,619

Notes: The table summarizes four variables capturing degree of vertical integration and four variables capturing the presence of station amenities. All variables are binary. Unweighted means and standard deviations (in parentheses) are calculated from station-level observations. "All" refers to all stations in the main analysis sample. "No Inc" indicates stations in municipalities too small to have income data reported. $Q1$ through $Q5$ denote quintiles of the municipality median income distribution. 'Vertically integrated' stations are those that are owned and operated by either a refiner or a wholesaler. 'Vertically independent' stations are not affiliated in any way with a refiner or wholesaler. 'Firm contract' denotes the firm sale of fuel to the retailer, and 'commission contract' indicates that the retailer earns a commission on sales above some pre-negotiated price.

Table A3: *Full-sample average tax pass-through, including País Vasco and Navarra*

	dCDH			TWFE
	(1)	(2)	(3)	(4)
Panel A. Sales-weighted				
Average pass-through	0.923	0.915	0.913	0.931
	(0.014)	(0.014)	(0.014)	(0.030)
N	24,380	45,630	63,535	449,482
Post-period	6	12	18	12
Panel B. Unweighted				
Average pass-through	0.871	0.895	0.899	0.939
	(0.006)	(0.006)	(0.009)	(0.025)
N	30,616	55,817	77,119	515,385
Post-period	6	12	18	12

Notes: Estimates are average pass-through rates of tax changes into retail prices (ρ; cent/liter price change per cent/liter tax change) obtained via Equations 5-7 in columns 1-3 ('dCDH') and via Equation 3 in column 5 ('TWFE'). The sample is the main analysis sample plus Navarra and País Vasco regions (for which income data are unavailable). An observation is a station-year-month. 'N' reports the number of treatment-group post-period observations used in columns 1-3 and the number of total observations used in column 4. The two Panels (A and B) show results with and without analytic weights equal to station-year diesel quantity sold, respectively. 'Post-period" gives the number of months included in estimation. Standard errors are clustered by province.

Table A4: *Average tax pass-through by income group, alternative clustering/observation levels*

	No Inc (1)	Q1 (2)	Q2 (3)	Q3 (4)	Q4 (5)	Q5 (6)
Panel A. Region-level clustering						
Average pass-through	0.868	0.848	0.958	0.935	0.943	0.954
	(0.020)	(0.011)	(0.016)	(0.010)	(0.013)	(0.017)
N	11,987	4,981	7,167	5,867	8,851	5,136
Panel B. Region-by-income-quintile monthly observations						
Average pass-through	0.871	0.857	0.927	0.936	0.944	0.951
	(0.011)	(0.012)	(0.016)	(0.012)	(0.012)	(0.004)
N	85	49	73	85	85	73

Notes: Estimates are average pass-through rates of tax changes into retail prices (ρ; cent/liter price change per cent/liter tax change) obtained via Equations 5-7, for each of six income groups. Each panel provides results with a different deviation from the base specification (whose results are presented in Table 3, Panel A), which uses station-year-month observations weighted by station-year diesel sales, median-income quintiles, a twelve-month post-period, and province-level clustering of standard errors. Panel A provides results using region-level clustering of standard errors and station (-year-month) observations. Panel B uses the same region-level clustering but region-quintile (-year-month) observations. Q1-Q5 indicate quintiles of the municipality median income distribution; "No Inc" denotes the group of stations in towns too small to be included in the data. 'N' reports the number of treatment-group post-period observations used.

Table A5: *Average tax pass-through by income group, alternative sample definitions*

	No Inc (1)	Q1 (2)	Q2 (3)	Q3 (4)	Q4 (5)	Q5 (6)
Panel A. Excluding regions where gas and diesel tax hikes have different sizes						
Average pass-through	0.855	0.925	0.959	0.960	0.947	0.970
	(0.059)	(0.014)	(0.022)	(0.017)	(0.026)	(0.007)
N	9,210	3,306	3,099	3,434	5,492	3,878
Panel B. Including stations near cross-border rivals						
Average pass-through	0.868	0.895	0.958	0.931	0.943	0.848
	(0.051)	(0.013)	(0.020)	(0.011)	(0.020)	(0.011)
N	12,126	5,017	7,432	5,919	8,858	4,981
Panel C. Only municipalities with six years of income data						
Average pass-through	0.868	0.837	0.963	0.938	0.952	0.956
	(0.051)	(0.014)	(0.022)	(0.010)	(0.025)	(0.018)
N	11,987	5,154	6,758	5,959	8,534	5,000

Notes: Estimates are average pass-through rates of tax changes into retail prices (ρ; cent/liter price change per cent/liter tax change) obtained via Equations 5-7, for each of six income groups. Each panel uses the base dCDH specification (Table 3, Panel A) with an alternative analysis sample. All panels use station-year-month observations weighted by station-year diesel sales, median-income quintiles, a twelve-month post-period, and province-level clustering of standard errors. Panel A drops from the sample three regions whose concurrent gasoline and diesel tax hikes are of different size. Panel B newly includes stations that are within ten minutes of a gas station in another region. Panel C omits municipalities without a full set of annual income measures from 2008-2013. Q1-Q5 indicate quintiles of the municipality median income distribution; "No Inc" denotes the group of stations in towns too small to be included in the data. 'N' reports the number of treatment-group post-period observations used.

Table A6: *Average tax pass-through by income group, alternative variable definitions*

	No Inc (1)	Q1 (2)	Q2 (3)	Q3 (4)	Q4 (5)	Q5 (6)
Panel A. Quintiles of 2010 median income						
Average pass-through	0.871	0.921	0.951	0.931	0.940	0.955
	(0.047)	(0.026)	(0.023)	(0.014)	(0.013)	(0.016)
N	12,217	4,701	6,346	5,890	9,260	5,545
Panel B. Weights by mean of station sales across years						
Average pass-through	0.866	0.836	0.958	0.938	0.944	0.961
	(0.048)	(0.012)	(0.017)	(0.014)	(0.015)	(0.014)
N	11,992	4,981	7,179	5,867	8,851	5,137

Notes: Estimates are average pass-through rates of tax changes into retail prices (ρ; cent/liter price change per cent/liter tax change) obtained via Equations 5-7, for each of six income groups. Both panels use station-year-month observations weighted by station-year diesel sales, median-income quintiles, a twelve-month post-period, and province-level clustering of standard errors. In Panel A, income quintiles are defined according to 2010 values, instead of 2008-2013 averages as in the base specification (Table 3, Panel A). In Panel B, sales weights are station annual means across years, rather than year-specific totals. Q1-Q5 indicate quintiles of the municipality median income distribution; "No Inc" denotes the group of stations in towns too small to be included in the data. 'N' reports the number of treatment-group post-period observations used.

Table A7: *Pass-through and brand concentration*

	All (1)	No Inc (2)	Q1 (3)	Q2 (4)	Q3 (5)	Q4 (6)	Q5 (7)
Panel A. Full brand saturation							
Average pass-through	0.937	0.927	1.003	0.943	0.909	1.024	0.964
	(0.010)	(0.014)	(0.032)	(0.030)	(0.016)	(0.023)	(0.013)
N	5,896	4,117	276	380	349	212	74
Panel B. Zero brand saturation							
Average pass-through	0.893	0.796	0.840	0.950	0.907	0.945	0.907
	(0.033)	(0.101)	(0.014)	(0.025)	(0.020)	(0.036)	(0.045)
N	16,433	4,813	2,699	2,906	2,000	2,586	1,236

Notes: Estimates are average pass-through rates of tax changes into retail prices (ρ; cent/liter price change per cent/liter tax change) obtained via Equation 7, for each of six income groups. In Panel A, only stations with no different-branded competitors within a ten-minute drive are included; in Panel B, only stations with competitors within a ten-minute drive and no shared branding whatsoever are considered. Column 1 displays overall average pass-through rates, and columns 2-7 display rates by income grouping (quintiles $Q1$-$Q5$ and the group of stations without income data). All specifications use station-year-month observations, weighted by station-year diesel quantity sold, with a twelve-month post-period and standard errors clustered by province. 'N' is the number of treatment-group post-period observations used.

Table A8: *Pass-through and sales volume*

	All (1)	No Inc (2)	Q1 (3)	Q2 (4)	Q3 (5)	Q4 (6)	Q5 (7)
Panel A. Bottom-quintile sales volume							
Average pass-through	0.958	0.929	1.003	0.961	1.062	1.012	0.983
	(0.013)	(0.017)	(0.040)	(0.029)	(0.103)	(0.020)	(0.035)
N	8,588	4,272	484	983	655	901	473
Panel B. Top-quintile sales volume							
Average pass-through	0.895	0.751	0.879	0.989	0.921	0.931	0.953
	(0.028)	(0.111)	(0.062)	(0.018)	(0.014)	(0.019)	(0.008)
N	8,880	1,507	296	1,369	1,334	2,440	1,448

Notes: Estimates are average pass-through rates of tax changes into retail prices (ρ; cent/liter price change per cent/liter tax change) obtained via Equation 7, for each of six income groups. In Panel A, only stations in the bottom quintile of the station sales volume distribution are included; in Panel B, only stations in the top quintile are considered. Column 1 displays overall average pass-through rates, and columns 2-7 display rates by income grouping (quintiles $Q1$-$Q5$ and the group of stations without income data). All specifications use station-year-month observations, weighted by station-year diesel quantity sold, with a twelve-month post-period and standard errors clustered by province. 'N' is the number of treatment-group post-period observations used.

Table A9: *Pass-through, vertical integration, and local competition*

	(1)	(2)	(3)	(4)	(5)
Panel A. Local competition	Q1	Q2	Q3	Q4	Q5
Average pass-through	0.916	0.978	0.867	0.934	0.927
	(0.015)	(0.024)	(0.059)	(0.009)	(0.014)
N	8,946	8,823	8,834	9,001	8,397
Panel B. Vertical integration	COCO	CODO-F	CODO-C	DODO-F	DODO-C
Average pass-through	0.928	0.987	0.962	0.952	0.969
	(0.011)	(0.025)	(0.008)	(0.015)	(0.011)
N	7,880	2,243	9,278	3,635	5,675

Notes: Estimates are average pass-through rates of tax changes into retail prices (ρ; cent/liter price change per cent/liter tax change) obtained via Equation 7, for each of five mutually exclusive subsamples. In Panel A, each column label denotes a quintile of the distribution of number of competing fuel stations within a ten-minute drive, where each competitor's contribution is inversely weighted by travel time in minutes. In Panel B, each column label denotes a different contract type, and only refiner-branded stations are used. COCO means (upstream) company owned, company operated – i.e., by a refiner or wholesaler. CODO indicates company owned, dealer (i.e., retailer) operated. DODO abbreviates dealer owned, dealer operated. Lastly, "F" and "C" suffixes denote "firm" sale of fuel to the retailer and "commission" contract (where the retailer earns a commission on sales above some pre-negotiated price), respectively. Column 1 displays overall average pass-through rates, and columns 2-7 display rates by income grouping (quintiles $Q1$-$Q5$ and the group of stations without income data). All specifications use station-year-month observations, weighted by station-year diesel quantity sold, with a twelve-month post-period and standard errors clustered by province. 'N' is the number of treatment-group post-period station-months used.